

1. What is a Computer Network?

A **computer network** is a group of two or more computers or devices connected together to share data, information, and resources. These devices communicate with each other through wired or wireless connections. A computer network helps users share files, printers, internet, and other services quickly and easily.

Key Points:

- Connects two or more computers.
 - Shares data and resources.
 - Uses wired or wireless connection.
 - Makes communication fast and easy.
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2. Discuss Various Types of Network Topologies

Network topology means the physical or logical arrangement of computers and cables in a network. It shows how devices are connected with each other.

The main types of network topologies are:

a) Bus Topology

In a bus topology, all computers are connected to one main cable called the **backbone cable**. Data travels through this cable to all devices.

b) Star Topology

In a star topology, every computer is connected to a central device such as a **switch** or **hub**. All communication passes through the central device.

c) Ring Topology

In a ring topology, each computer is connected to two neighboring computers, forming a circular path. Data moves around the ring until it reaches the destination.

d) Mesh Topology

In a mesh topology, every computer is directly connected to every other computer. This provides multiple paths for data transmission.

e) Tree Topology

A tree topology combines the features of **star** and **bus** topologies. It has a hierarchical structure with one main line and several branches.

3. Advantages and Disadvantages of Each Topology

Topology	Advantages	Disadvantages
Bus	Low cost, easy to install, uses less cable.	Backbone cable failure stops the whole network. Performance becomes slow with many devices.
Star	Easy to manage, easy to add devices, failure of one cable does not affect others.	If the central hub/switch fails, the whole network stops. Uses more cable.
Ring	Data flows in an orderly way. No data collision.	If one device or cable fails, the whole network may stop. Difficult to troubleshoot.
Mesh	Very reliable, multiple paths for data, high security.	Very expensive and difficult to install because many cables are needed.
Tree	Easy to expand, suitable for large networks, easy management.	Backbone failure affects many devices. Installation cost is higher.

4. What are the Applications of Computer Networks?

Computer networks are used to share information and resources between users and devices. They make communication faster, easier, and more efficient.

Applications of Computer Networks:

- File sharing between computers.
- Sharing printers, scanners, and other devices.
- Internet access.
- E-mail and instant messaging.
- Video and voice calls.
- Online learning (E-learning).
- Online banking and shopping.
- Business communication and data sharing.

5. What is OSI Model? Explain the Functions, Protocols and Services of Each Layer?

The **OSI (Open Systems Interconnection) Model** is a standard model used to explain how data moves from one computer to another through a network. It has **7 layers**, and each layer performs a specific job.

Layer	Function	Example Protocols	Service
7. Application	Provides network services to users.	HTTP, FTP, SMTP, DNS	User communication
6. Presentation	Converts, encrypts and compresses data.	SSL, TLS	Data formatting
5. Session	Creates, manages and ends communication sessions.	NetBIOS, RPC	Session management
4. Transport	Delivers data reliably and controls errors.	TCP, UDP	End-to-end delivery
3. Network	Finds the best path and routes data.	IP, ICMP	Routing
2. Data Link	Sends data between two connected devices and detects errors.	Ethernet, PPP, HDLC	Frame delivery
1. Physical	Sends bits through cables or wireless media.	Ethernet, USB	Bit transmission

Easy way to remember:

Application → Presentation → Session → Transport → Network → Data Link → Physical

6. Explain the Following

I. LAN (Local Area Network)

A **LAN** is a network that connects computers within a **small area**, such as a room, office, school, or building. It provides high-speed communication and resource sharing.

Key Points:

- Covers a small area.
 - High speed.
 - Low cost.
 - Example: School computer lab.
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II. MAN (Metropolitan Area Network)

A **MAN** connects multiple LANs within a **city or metropolitan area**. It is larger than a LAN but smaller than a WAN.

Key Points:

- Covers a city.
 - Connects many LANs.
 - Medium speed.
 - Example: A network connecting different university campuses in one city.
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III. WAN (Wide Area Network)

A **WAN** connects computers over a **very large area**, such as different cities or countries. It is used to connect remote offices and users. The **Internet** is the largest example of a WAN.

Key Points:

- Covers a large area.
- Connects different cities or countries.
- Uses public or private communication links.
- Example: The Internet.

7. What are the network components? What do you mean by client-server network?

Answer

A computer network relies on several basic hardware and software parts to function properly. The primary network components include:

- **Physical Media:** Cables (Twisted pair, Fiber optic) or wireless signals (Wi-Fi) that carry data.
- **Interconnecting Devices:** Hardware like Hubs, Switches, and Routers that connect devices together.
- **Computers/Nodes:** Devices attached to the network, functioning as Clients or Servers.
- **Software & Applications:** Operating systems, protocols, and programs like web browsers and email clients.

A **Client-Server Network** is a network architecture where roles are divided between two types of computers. The **Server** is a powerful central computer that stays running 24/7 to store data, manage resources, and provide services. The **Client** is a user's computer (like a laptop or phone) that requests services or files from the server. For example, when watching a video on YouTube, your phone acts as the client requesting data, and Google's computer acts as the server.

8. Write some protocol names of each layer of TCP/IP model.

Answer

The TCP/IP model organizes communication protocols into four distinct layers. Each layer performs a specific role and uses dedicated protocols:

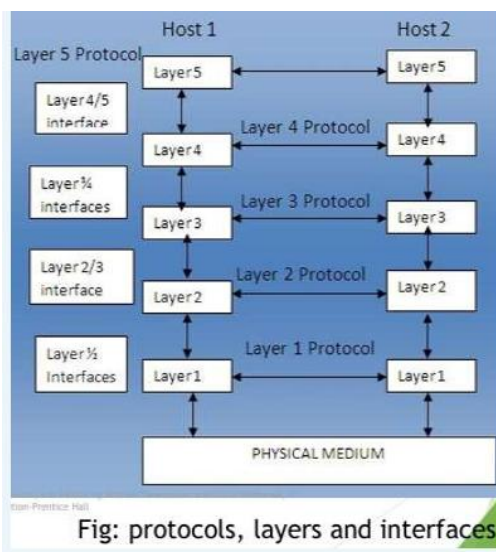
Layer Name PDF	Key Protocols Used PDF
1. Application Layer	HTTP, HTTPS, FTP, SMTP, DNS, DHCP
2. Transport Layer	TCP (Transmission Control Protocol), UDP (User Datagram Protocol)
3. Internet Layer	IP (IPv4, IPv6), ICMP, ARP, RARP
4. Network Access / Link Layer	Ethernet, PPP, HDLC, Wi-Fi (802.11)

9. Explain protocol, layer, peer and interface using layered diagram.

Answer

To simplify complex network communication, systems are structured using a layered model. Four fundamental terms define this architecture:

- **Layer:** A specific level in the network hierarchy that performs one dedicated task. Higher layers rely on the services provided by lower layers.
- **Protocol:** A strict set of rules that governs how two devices communicate with each other.
- **Peer:** Entities or processes that operate at the exact same layer on different devices. They communicate logically using a common protocol.
- **Interface:** The boundary between two adjacent layers on the same machine that defines what services the lower layer offers to the upper layer.



10. Explain connection multiplexing, connectionless and connection-oriented services.

Answer

Computer networks provide different modes of service to transfer data efficiently:

- **Connection-Oriented Service:** A dedicated connection is established first between the sender and receiver before any data transfer begins. Once communication ends, the connection is released. It provides reliable delivery.
 - *Example:* TCP protocol (similar to a phone call).
- **Connectionless Service:** Data packets are sent directly to the destination without setting up a connection beforehand. Each packet travels independently.
 - *Example:* UDP protocol (similar to sending a letter by post).
- **Connection Multiplexing:** The technique of combining multiple high-layer connections into a single lower-layer connection (Upward Multiplexing), or splitting one high-layer connection across multiple lower-layer connections (Downward Multiplexing) to optimize network resources.

11. Explain OSI and TCP/IP model

Answer

The **OSI (Open Systems Interconnection) Model** is a 7-layer theoretical framework developed by ISO to standardize network functions:

1. **Application Layer:** User interface and network services (HTTP, FTP).
2. **Presentation Layer:** Data formatting, encryption, and compression.
3. **Session Layer:** Manages and synchronizes communication sessions.
4. **Transport Layer:** Ensures end-to-end reliable delivery (TCP, UDP).
5. **Network Layer:** Routing and logical IP addressing (IP, Router).
6. **Data Link Layer:** Framing, MAC addressing, and error detection (Ethernet, Switch).
7. **Physical Layer:** Bit transmission over physical media (Cables, Hub).

The **TCP/IP Model** is the practical 4-layer architecture used by the real-world Internet. It combines several OSI layers together:

TCP/IP Layer Name PDF	Equivalent OSI Layers PDF	Key Protocols PDF
1. Application Layer	Application, Presentation, Session (Layers 7, 6, 5)	HTTP, DNS, SMTP
2. Transport Layer	Transport (Layer 4)	TCP, UDP
3. Internet Layer	Network (Layer 3)	IP, ICMP, ARP

4. Network Access Layer	Data Link, Physical (Layers 2, 1)	Ethernet, Wi-Fi, PPP
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12. Describe HDLC frame format.

Answer

HDLC (High-Level Data Link Control) is a bit-oriented Data Link Layer protocol that packages data into synchronous frames for transmission.



The HDLC frame consists of six fields:

- **Start Flag (8 bits):** Contains the fixed bit pattern 01111110 to mark the beginning of the frame.
- **Address Field (8 or more bits):** Identifies the destination station receiving the frame.
- **Control Field (8 or 16 bits):** Carries control info, sequence numbers, and acknowledgments for flow and error control.
- **Information Field (Variable length):** Carries the actual user data payload.
- **FCS / Frame Check Sequence (16 or 32 bits):** Contains CRC error-checking code to detect transmission errors.
- **End Flag (8 bits):** Same bit pattern 01111110 to mark the end of the frame.

13. How many types of HDLC frame there are? Explain control field formation of each of those.

Answer

There are **three types** of HDLC frames based on the structure and purpose of their Control Field:

1. **Information Frame (I-Frame):**
 - **Purpose:** Used to transfer user data and piggybacked acknowledgments.
 - **Control Field Structure:** The first bit is always 0. It contains a 3-bit Sequence Number (SEQ) for the frame being sent, a 1-bit Poll/Final (P/F) bit, and a 3-bit Next Expected Frame Number (NEXT) for acknowledgment.
2. **Supervisory Frame (S-Frame):**
 - **Purpose:** Used for flow and error control (carries no user data).

- **Control Field Structure:** The first two bits are 10. It includes a 2-bit Type field (RR, REJ, RNR, SREJ), a P/F bit, and a 3-bit NEXT expected frame number.

3. Unnumbered Frame (U-Frame):

- **Purpose:** Used for link management, setting modes, and connectionless data transfer.
- **Control Field Structure:** The first two bits are 11. It contains a P/F bit and 5 modifier bits to specify commands like DISC (Disconnect) or SABM (Set Asynchronous Balanced Mode).

14. When and why we use point –to-point connection?

Answer

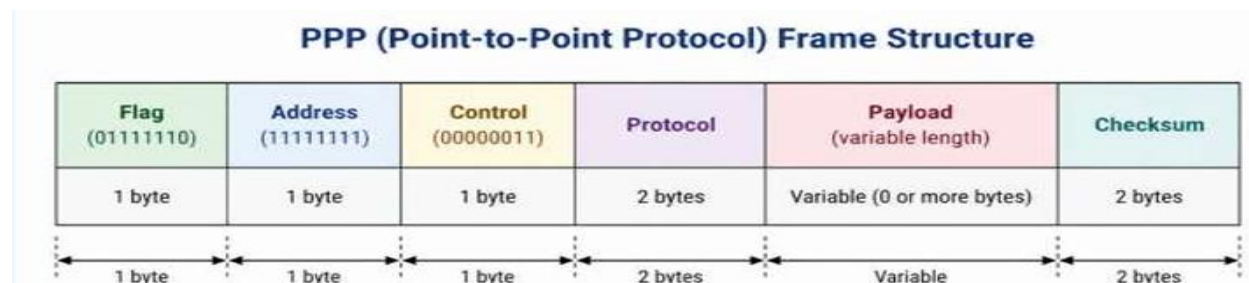
A **Point-to-Point Connection** is a dedicated communication link established directly between two specific network devices or routers.

- **When we use it:**
 1. When connecting two LAN networks over a long geographical distance (LAN-to-LAN connection via a leased line).
 2. When a home user or remote user connects directly to an Internet Service Provider (ISP) using a dial-up modem or broadband connection.
- **Why we use it:**
 - **Dedicated Bandwidth:** The entire link bandwidth is reserved exclusively for communication between those two devices, ensuring fast and consistent speeds.
 - **No Collisions:** Since only two devices share the line, data signals never collide with other traffic.
 - **Simplicity & Security:** It does not require complex routing or multi-device addressing, making data transfer simple, private, and secure.

15. Draw the ppp frame format. Explain each field.

Answer

PPP (Point-to-Point Protocol) uses an HDLC-like frame format to deliver data over direct point-to-point links.



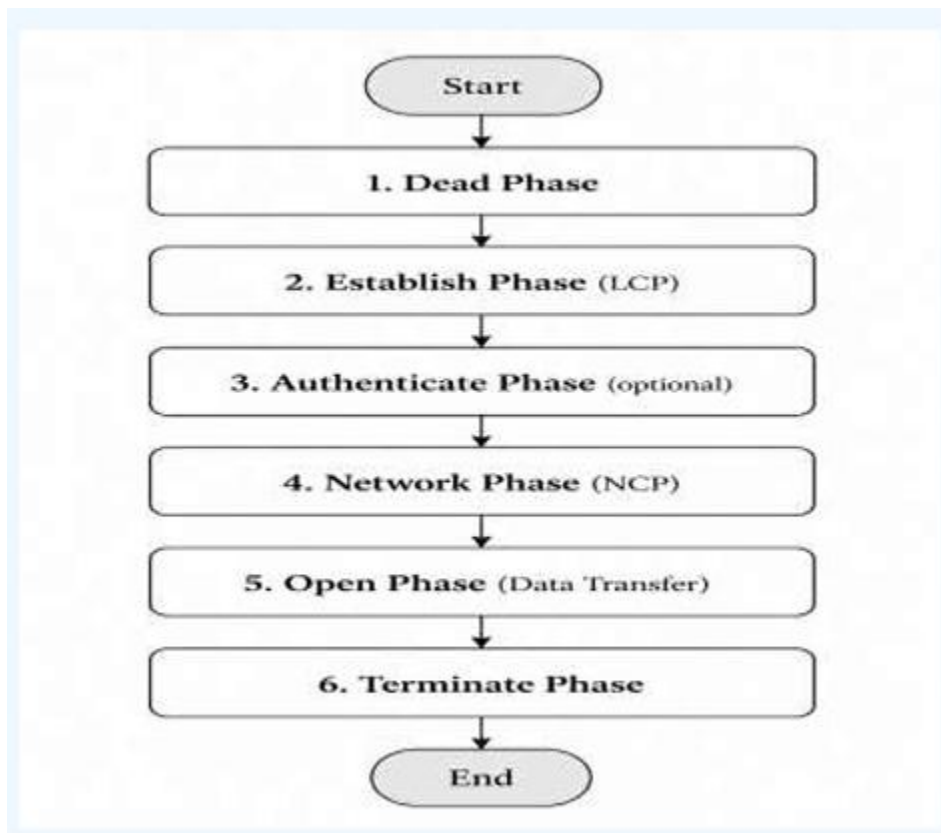
Field explanations:

- **Flag (1 byte):** Set to 01111110 to mark the beginning and end of the PPP frame.
- **Address (1 byte):** Always set to 11111111 (Broadcast) because point-to-point links do not require individual device addresses.
- **Control (1 byte):** Always set to 00000011 indicating Unnumbered Mode (no default sequence numbers or acknowledgments).
- **Protocol (2 bytes):** Identifies the type of data inside the payload. If it starts with bit 0, it carries network layer data (e.g., IP); if it starts with bit 1, it carries control messages (e.g., LCP or NCP).
- **Payload (Variable):** The actual user data or control data being transmitted (default up to 1500 bytes).
- **Checksum (2 bytes):** Used for error detection to ensure the frame arrived undamaged.

16. Explain the diagram for bring a line up and down in a PPP communication.

Answer

A PPP connection goes through six distinct sequential phases (states) from establishment to termination:



Explanation of phases:

1. **Dead Phase:** The physical line is idle and no active connection exists.
2. **Establish Phase:** LCP (Link Control Protocol) opens the connection and negotiates link configuration options.
3. **Authenticate Phase (Optional):** Both sides verify each other's identity using authentication protocols (like PAP or CHAP).
4. **Network Phase:** NCP (Network Control Protocol) configures network-layer parameters (such as IP address assignment).
5. **Open Phase:** Data transfer takes place actively between the devices.
6. **Terminate Phase:** LCP closes the connection, releases resources, and drops the line back to the Dead phase.

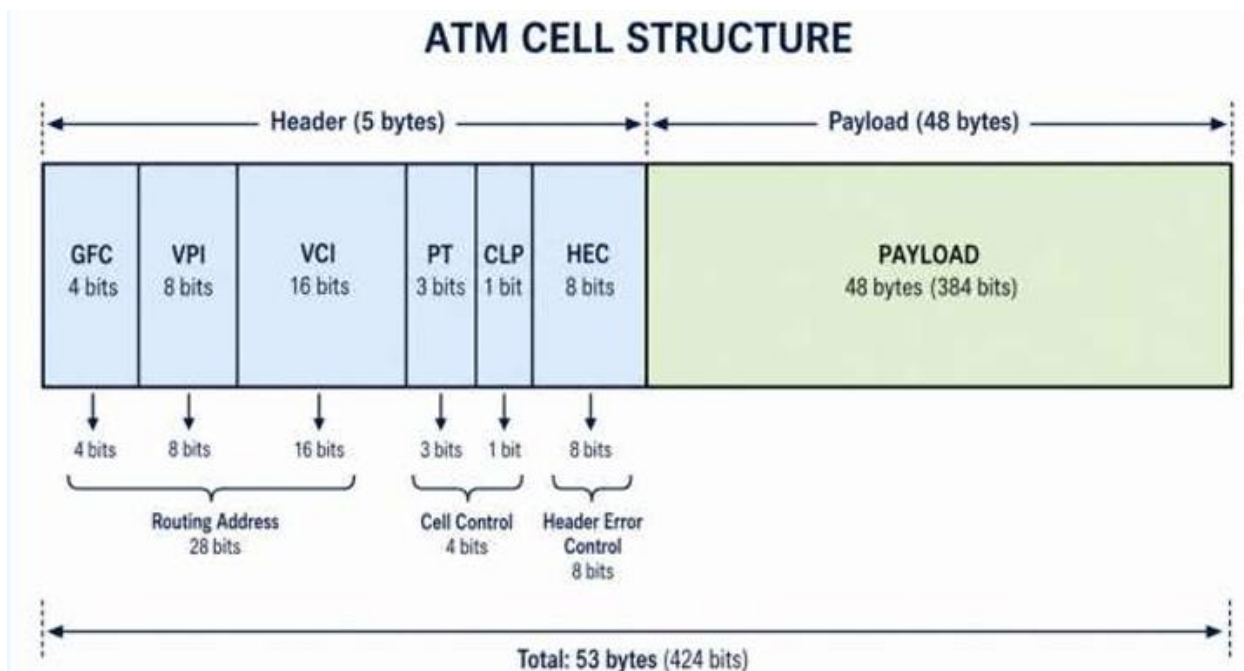
17. ATM has how many cell formats? Explain the commonly used one.

Answer

ATM (Asynchronous Transfer Mode) uses **two** cell formats depending on where the cell is transmitted:

1. **UNI (User-Network Interface):** Used between a user device (like a computer) and the ATM switch.
2. **NNI (Network-Network Interface):** Used between two ATM switches inside the network.

The most commonly used format is the **UNI (User-Network Interface)** format. An ATM cell has a fixed size of **53 bytes** (5-byte header + 48-byte payload).



UNI Header Fields (Easy Explanation)

- **GFC (Generic Flow Control - 4 bits):** Controls the flow of data between the **user** and the **network**.
- **VPI (Virtual Path Identifier - 8 bits):** Identifies the **virtual path** that the ATM cell follows.
- **VCI (Virtual Channel Identifier - 16 bits):** Identifies the **virtual channel** inside the virtual path. **VPI and VCI together** help route the cell to the correct destination.
- **PT (Payload Type - 3 bits):** Indicates the **type of data** in the cell, such as user data or control/management data.
- **CLP (Cell Loss Priority - 1 bit):** Shows the **priority** of the cell. If **CLP = 1**, the cell has **low priority** and may be dropped during network congestion.
- **HEC (Header Error Control - 8 bits):** Checks the ATM **header for errors** and helps identify the start of each cell.

18. Why Standardization is necessary? Explain with example

Answer

Standardization is necessary to establish common rules, protocols, and hardware specifications across the telecommunication industry.

- **Why it is necessary:**
 - **Interoperability:** Ensures that equipment manufactured by different companies can communicate seamlessly.
 - **Cost Reduction:** Enables mass production of hardware, which lowers manufacturing costs and consumer prices.
 - **Global Adoption:** Encourages widespread adoption of new networking technologies.
- **Example:** Before network standards, if Company A made a computer and Company B made a router, they could not talk to each other because each used proprietary technologies. When the **IEEE** introduced standard protocols like **IEEE 802.3 (Ethernet)**, devices from any manufacturer (e.g., HP, Dell, Cisco) could connect and communicate out of the box.

19. How does CSMA/CD work? Explain

Answer

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) is a media access control method used in Ethernet (bus topology) to share a common communication cable.

How it works step-by-step:

1. **Carrier Sense (Listen):** A station checks the physical cable to see if another device is transmitting.
2. **Multiple Access (Transmit):** If the line is free, the station begins transmitting its data. If busy, it waits.

3. **Collision Detection (Detect):** While transmitting, the station listens to the cable. If two stations transmit simultaneously, their signals collide.
4. **Jam Signal & Backoff (Recover):** When a collision is detected, both stations stop transmitting immediately. Each station waits for a random period (backoff time) before trying to retransmit.

20. What do mean by token? Explain IEEE 802.4 and 802.5.

Answer

A **Token** is a special, small control frame that continuously circulates around a network. Only the station currently holding the free token is authorized to transmit data. This completely eliminates collisions.

- **IEEE 802.4 (Token Bus):**
 - Devices are physically connected in a **bus topology**, but the token is passed logically in a **circle (ring)** from node to node.
 - The node holding the token attaches data to it and sends it along the bus.
- **IEEE 802.5 (Token Ring):**
 - Devices are physically connected in a **ring topology**.
 - The token travels around the physical ring from one workstation to the next. Each device inspects or passes the token until it reaches its destination.

21. Why not just one big LAN?

Answer

Building a single, massive LAN for an entire organization creates severe performance and operational issues:

- **Large Collision Domain:** More connected devices lead to frequent collisions, forcing constant retransmissions and degrading speed.
- **Shared Bandwidth Limits:** All stations share the same physical cable capacity, reducing overall network performance.
- **Cable Length Constraints:** Physical standards (like Ethernet) have maximum cable length limits (e.g., 100 meters for UTP), making a single LAN impossible over long distances.
- **High Delay & Congestion:** Token-passing or broadcast networks slow down significantly when too many stations wait for turns or receive broadcast packets.

22. What devices are available for LAN's interconnection? Mention which layer they work.

Different networking devices are used to connect LANs or LAN segments. Each device works at a different **OSI layer**.

Device	OSI Layer	Data Handled
Repeater / Hub	Layer 1 (Physical)	Bits / Electrical Signals
Bridge / Switch	Layer 2 (Data Link)	Frames (MAC Address)
Router	Layer 3 (Network)	Packets (IP Address)
Application Gateway	Layer 7 (Application)	Application Data

23. Explain Hub and Bridge. Compare Them.

Hub

A **Hub** is a **Layer 1 (Physical Layer)** device. It connects multiple computers in a network. When it receives data from one computer, it **sends the data to all other ports**. It does not check any address.

Bridge

A **Bridge** is a **Layer 2 (Data Link Layer)** device. It connects two or more LAN segments. It checks the **MAC address** and sends data **only to the correct destination**.

Comparison

Feature	Hub	Bridge
OSI Layer	Layer 1	Layer 2
Forwarding	Sends to all ports	Sends only to the correct port
Address Check	No	Yes (MAC Address)
Collision Domain	One collision domain	Separate collision domain for each port
Device Type	Simple (Dumb)	Intelligent (Smart)

24. Give a Bridge Learning Example.

A **bridge** learns the **MAC address** of devices automatically. This is called **self-learning**.

Step 1: C sends data to D

- Host **C** sends a frame to **D**.
- The bridge **does not know** where D is.

- So, it **floods** the frame to all ports.
- At the same time, it learns:
 - **C → Port 1**

Step 2: D replies to C

- Host **D** sends a reply.
- The bridge sees the frame comes from **Port 2**.
- It learns:
 - **D → Port 2**

Step 3: Future Communication

- Now the bridge knows both devices.
- It sends frames **directly** between **Port 1** and **Port 2**.
- **No flooding** is needed.

Filtering Table

MAC Address	Port
C	Port 1
D	Port 2

25. What will happen if there is a loop between two nodes?

If there is a **loop** between two LAN segments, the network faces several problems.

Problems:

- **Broadcast Storm:** Broadcast frames keep circulating, using all network bandwidth.
- **Frame Duplication:** The same frame is copied and delivered many times.
- **MAC Table Instability:** The bridge keeps changing MAC address entries, causing incorrect forwarding.

As a result, the network becomes **slow or may stop working**.

26. What is Spanning Tree? How does it help Bridge?

Spanning Tree Protocol (STP) is a **Layer 2 protocol (IEEE 802.1D)**. It creates a **loop-free network** by blocking extra paths.

How STP Helps a Bridge

- Removes loops by blocking unnecessary links.
- Prevents broadcast storms.
- Prevents frame duplication.
- Keeps the MAC table stable.
- If the main link fails, it automatically activates the backup link to keep the network running.

27. Compare and contrast between routers and bridge.

Answer

Bridges and Routers both forward network traffic, but they operate at different layers of the OSI model.

Similarities:

- Both are store-and-forward devices used to connect networks or network segments together.

Differences (Comparison Table):

Feature PDF	Bridge PDF	Router PDF
OSI Layer	Layer 2 (Data Link Layer)	Layer 3 (Network Layer)
Addressing Used	Hardware MAC Address	Logical IP Address
Forwarding Table	Filtering / MAC Table	Routing Table
Broadcast Protection	Forwards all broadcasts (No protection)	Blocks broadcasts (Provides firewall/isolation)
Loop Prevention	Spanning Tree Protocol (STP)	TTL (Time to Live) counter & Routing Algorithms
Best For	Small networks (up to ~200 hosts)	Large networks (thousands of hosts)

28. What do you mean by cut-through switching?

Answer

Cut-Through Switching is an advanced frame-forwarding technique used by network switches.

- Instead of waiting for the entire data frame to arrive in memory (Store-and-Forward), the switch reads only the first few bytes containing the **destination MAC address** and immediately begins forwarding the frame to the target port.
- **Key Advantage:** It significantly reduces end-to-end delay (latency).
- **Use Case:** Ideal for real-time applications requiring extremely low latency.

Plaintext

[Read Destination MAC Address (First 6 Bytes)] ---> [Immediately Forward Frame]

29. Compare and contrast between switch and bridge.

Answer

Switches and bridges are fundamentally similar as both make forwarding decisions based on Layer 2 MAC addresses, but they differ in scale and operation.

Similarities:

- Both operate at Layer 2 (Data Link Layer).
- Both use MAC addresses for forwarding and isolate collision domains per port.

Differences (Comparison Table):

Feature PDF	Bridge PDF	Switch PDF
Primary Purpose	Connects whole LAN segments	Connects individual host computers
Number of Ports	Few ports (typically 4 to 8)	Many ports (12, 24, 48+ ports)
Simultaneous Links	Limited parallel transmission	Supports multiple simultaneous connections
Hardware	Software-based processing	Hardware/ASIC-based fast processing
Buffer Memory	Per line card	Dedicated buffer space per port

30. What is FDDI? How does it work?

Answer

FDDI (Fiber Distributed Data Interface) is a high-speed LAN standard that transmits data over optical fiber cables at **100 Mbps** across distances up to **200 kilometers**.

How it works:

1. **Dual Ring Architecture:** It uses two counter-rotating fiber rings — a **Primary Ring** for active data transmission and a **Secondary Ring** for backup.
2. **Timed Token Passing:** A special token frame circulates around the ring. Only the station holding the token can transmit data.
3. **Self-Healing / Fault Tolerance:** If a cable breaks or a device fails, the primary and secondary rings wrap around automatically to form a single working ring.

31. What are the advantages of FDDI?

Answer

The key advantages of FDDI include:

- **High Speed & Throughput:** Provides 100 Mbps transmission speed (and up to 200 Mbps if both rings are used simultaneously).
- **High Fault Tolerance:** The dual-ring structure automatically reroutes traffic if a fiber cable snaps or a station fails.
- **Long Distance Coverage:** Can extend up to 200 km, supporting thousands of users across large campus networks or MANs.
- **Large Frame Size:** Supports a maximum frame size of **4,352 bytes** (much larger than Ethernet's 1,500 bytes), allowing better data transfer efficiency.
- **No Collisions:** Token passing guarantees orderly access without packet collisions.

32. Draw the FDDI frame formation.

Answer

An **FDDI Frame** consists of the following sequential fields:

PA	SD	FC	DA	SA	PDU	FCS	ED/FS
16 bits	8 bits	8 bits	48 bits	48 bits	up to 4478x8 bits	32 bits	16 bits

Figure 8.5: **FDDI** Frame Format — each field serves a specific role in frame identification, addressing, data, error checking, and status signalling.

Field Explanations:

- **PA (Preamble - 16 bits):** Synchronizes the receiver with incoming signals.

- **SD (Start Delimiter - 8 bits):** Marks the beginning of the frame.
- **FC (Frame Control - 8 bits):** Identifies the frame type (data or token).
- **DA (Destination Address - 48 bits):** MAC address of the target recipient.
- **SA (Source Address - 48 bits):** MAC address of the sender.
- **PDU (Protocol Data Unit):** The actual user data payload being carried.
- **FCS (Frame Check Sequence - 32 bits):** CRC checksum for error detection.
- **ED/FS (End Delimiter / Frame Status - 16 bits):** Marks frame termination and indicates if the frame was received/copied correctly.

33. Differentiate between Ethernet and token ring.

Answer

Ethernet and Token Ring are two different media access control methods used in local area networks.

Feature PDF	Ethernet (CSMA/CD) PDF	Token Ring (Token Passing) PDF
IEEE Standard	IEEE 802.3	IEEE 802.5
Topology	Bus / Star	Ring
Access Method	Contention-based (CSMA/CD)	Token Passing
Collisions	Yes, collisions can occur and are detected	No collisions (only token holder transmits)
Low Traffic Performance	Excellent (Fast, no waiting time)	Good (slight overhead from token passing)
High Traffic Performance	Degrades sharply due to collisions	Excellent and predictable (No collisions)

34. If there is a network like 172.168.0.0/17, then determine the followings:

Initial Information:

- **Given Network:** 172.168.0.0/17 (Class B Network)
- Default Class B Prefix = /16
- Default Host Bits = $32 - 16 = 16$ bits

a. How many subnets are possible?

- Borrowed Subnet Bits (n) = $17 - 16 = 1$ bit

- **Possible Subnets** = $2^n = 2^1 = 2$ subnets

b. How many hosts are there?

- Remaining Host Bits (h) = $32 - 17 = 15$ bits
- **Total Hosts per subnet** = $2^{15} = 32,768$
- **Usable Hosts per subnet** = $2^{15} - 2 = 32,766$ hosts

c. What is the subnet mask of the network?

- Binary Mask: 11111111.11111111.10000000.00000000
- **Dotted Decimal Subnet Mask** = **255.255.128.0**

d. In the 1st subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?

- **Network Address:** 172.168.0.0
- **1st Host Address:** 172.168.0.1
- **Last Host Address:** 172.168.127.254
- **Broadcast Address:** 172.168.127.255

e. In the 18th subnet, what are the IP addresses of network address, 1st host address, last host address and broadcast address?

- **Answer:** Since the prefix /17 borrows only 1 bit ($2^1 = 2$), **only 2 subnets are possible in total** (172.168.0.0/17 and 172.168.128.0/17). Therefore, **an 18th subnet does not exist** for this prefix.

35 & 37. Physical media available for Ethernet, Fast Ethernet, and Gigabit Ethernet and their maximum segment length.

Answer

The physical media types and maximum segment lengths for Ethernet standards are summarized below:

1. Classic Ethernet (10 Mbps):

- **10BASE5:** Thick Coaxial Cable | Max Length = **500 m**
- **10BASE2:** Thin Coaxial Cable | Max Length = **185 m**
- **10BASE-T:** UTP Cat 3/5 Copper Cable | Max Length = **100 m**
- **10BASE-FL:** Optical Fiber | Max Length = **2,000 m**

2. Fast Ethernet (100 Mbps):

- **100BASE-TX:** UTP Cat 5e Copper Cable | Max Length = **100 m**

- **100BASE-T4:** UTP Cat 3 Copper Cable (4 pairs) | Max Length = **100 m**
- **100BASE-FX:** Multimode Fiber = **2,000 m (Full Duplex)** / Singlemode Fiber = **20 km**

3. Gigabit Ethernet (1000 Mbps / 1 Gbps):

- **1000BASE-T:** UTP Cat 5e/6 Copper Cable | Max Length = **100 m**
- **1000BASE-SX:** Short Wavelength Multimode Fiber | Max Length = **275 m**
- **1000BASE-LX:** Long Wavelength Fiber (Singlemode) | Max Length = **5,000 m (5 km)**
- **1000BASE-LH:** Long Haul Singlemode Fiber | Max Length = **80 km**
- **1000BASE-CX:** Shielded Copper Twinax | Max Length = **25 m**

36. What leads to fast Ethernet?

Answer

Classic Ethernet running at 10 Mbps became too slow as file sizes increased and more computers joined local networks. This demand for higher speed led to the development of Fast Ethernet:

1. **Failure of Early Competitors:** Early high-speed optical ring technologies like **FDDI** and **Fiber Channel** failed to become popular for desktop systems due to complex management, expensive hardware chips, and high deployment costs.
2. **Reuse of CSMA/CD & Frame Format:** Instead of inventing a completely new protocol, engineers upgraded Ethernet to **100 Mbps** while retaining the simple, cheap, and familiar **CSMA/CD access protocol** and existing Ethernet frame format.
3. **Smooth Upgrade Path:** Keeping the same software interface allowed organizations to upgrade network speeds easily without replacing their entire existing infrastructure.

What are the physical media available for Ethernet, Fast Ethernet and Gigabit Ethernet? Also note their maximum segment length.

Ethernet Type	Standard	Medium Used	Max Segment Length
Classic Ethernet (10 Mbps)	10Base5	Thick Coaxial Cable	500 m
	10Base2	Thin Coaxial Cable	185 m
	10BaseT	UTP Cat 3/5 Copper	100 m
Fast Ethernet (100 Mbps)	100BaseTX	UTP Cat 5e Copper	100 m
	100BaseFX (MM)	Multimode Fiber	2 km
	100BaseFX (SM)	Single-mode Fiber	20 km
Gigabit Ethernet (1000 Mbps)	1000BaseT	UTP Cat 5e/6 Copper	100 m
	1000BaseSX	Multimode Fiber (Short-reach)	275 m
	1000BaseLX (MM)	Multimode Fiber	512 m
	1000BaseLX (SM)	Single-mode Fiber (Long-reach)	20 km
	1000BaseLH	Single-mode Fiber (Long-haul)	80 km

What leads to Fast Ethernet?

The transition and development that led to **Fast Ethernet** can be summarized through the following sequential points based on your course notes:

- **Sustained Growth and Network Crowding:** In the early days of networking, Classic Ethernet (operating at 10 Mbps) offered sufficient speeds. However, as more workstations and computers were connected to Local Area Networks (LANs) and users began exchanging significantly larger files, the original 10 Mbps bandwidth became a major bottleneck.
- **Failure of Ring-Based Optical Competitors:** Before Fast Ethernet was standardized, companies attempted to solve the speed crisis by designing faster, specialized network architectures based on ring-shaped optical cables, such as **FDDI** (Fiber Distributed Data Interface) and **Fiber Channel**. While they successfully utilized light signals to transmit data quickly, they completely failed to achieve mainstream popularity for desktop computers due to three primary barriers:
 1. **Complex Network Management:** Managing and troubleshooting the ring infrastructure was highly complicated.
 2. **High Chip Complexity:** The integrated circuits and chips required to run these protocols were overly complex to design.
 3. **Exorbitant Cost:** The specialized equipment was far too expensive for widespread commercial deployment.
- **The Strategic Choice to Upgrade Ethernet:** Because of the failures and high costs of FDDI and Fiber Channel, the networking industry collectively decided to improve the existing Ethernet standard rather than adopting an entirely new, complex technology.
- **Preservation of the Core Architecture:** This design philosophy directly resulted in **Fast Ethernet (100 Mbps)**. It succeeded because it intentionally retained Classic Ethernet's simple, inexpensive, and well-understood foundational rules—specifically reusing the exact same **CSMA/CD** (Carrier Sense Multiple Access with Collision Detection) media access protocol and the same frame/packet format. This allowed organizations to scale their bandwidth tenfold while fully preserving their existing software systems and engineering experience.

What are the Ethernet, Fast Ethernet and Gigabit Ethernet physical media types? Write down their maximum segment length.

1. Classic Ethernet (10 Mbps)

Classic Ethernet operates at 10 Mbps and uses various coaxial, twisted-pair, and fiber optic options:

- **10Base5:** Uses **Thick coaxial cable** in a bus topology with a maximum segment length of **500 meters**.
- **10Base2:** Uses **Thin coaxial cable** in a bus topology with a maximum segment length of **185 meters**.
- **10BaseT:** Uses **UTP Category 3/5 copper cable** in a tree (star) topology with a maximum segment length of **100 meters**.
- **10BaseFL:** Uses **Multimode or Single-mode fiber optic links** in a tree topology.

2. Fast Ethernet (100 Mbps)

Fast Ethernet upgrades the bandwidth to 100 Mbps using copper and fiber media:

- **100BaseTX:** Uses **UTP Category 5e copper cable** with a maximum segment length of **100 meters**.
- **100BaseFX (Multimode):** Uses **Multimode fiber** optic cable with a maximum segment length of **2 kilometers (2 km)**.
- **100BaseFX (Single-mode):** Uses **Single-mode fiber** optic cable with a maximum segment length of **20 kilometers (20 km)**.

3. Gigabit Ethernet (1000 Mbps / 1 Gbps)

Gigabit Ethernet provides 1 Gbps of bandwidth across copper, short-reach fiber, and long-reach fiber standards:

- **1000BaseT:** Uses **UTP Category 5e or Category 6 copper cable** with a maximum segment length of **100 meters**.
- **1000BaseSX:** Uses **Multimode fiber** (short reach) with a maximum segment length of **275 meters**.
- **1000BaseLX (Multimode):** Uses **Multimode fiber** (long reach) with a maximum segment length of **512 meters**.
- **1000BaseLX (Single-mode):** Uses **Single-mode fiber** (long reach) with a maximum segment length of **20 kilometers (20 km)**.
- **1000BaseLH:** Uses **Single-mode fiber** (long haul) with a maximum segment length of **80 kilometers (80 km)**.

Ethernet Generation	Standard	Media Name	Physical Media / Cable Type Used	Maximum Segment Length
Classic Ethernet (10 Mbps)		10Base5	Thick coaxial cable	500 m
		10Base2	Thin coaxial cable	185 m
		10BaseT	UTP Cat 3/5 copper cable	100 m
		10BaseFL	Multimode/Single-mode fiber	Fiber optic link
Fast Ethernet (100 Mbps)		100BaseTX	UTP Cat 5e copper	100 m
		100BaseFX (MM)	Multimode fiber	2 km
		100BaseFX (SM)	Single-mode fiber	20 km
Gigabit Ethernet (1000 Mbps)		1000BaseT	UTP Cat 5e/Cat 6 copper	100 m
		1000BaseSX	Multimode fiber	275 m
		1000BaseLX (MM)	Multimode fiber	512 m
		1000BaseLX (SM)	Single-mode fiber	20 km
		1000BaseLH	Single-mode fiber	80 km

Explain Non-adaptive & Adaptive routing algorithm.

Routing algorithms are broadly categorized into two main types based on how they make their decisions: Non-Adaptive (Static) Routing and Adaptive (Dynamic) Routing.

1. Non-Adaptive (Static) Routing

Definition

Non-adaptive routing algorithms, commonly known as static routing, choose paths and routing decisions in advance. They do not factor in the current live traffic or network topology changes while the network is actively running.

How it Works

- **Pre-calculated Offline:** The optimal route between every pair of routers is calculated beforehand, completely offline. This is typically done by a network administrator or a specialized network planning tool.
- **Boot-time Loading:** These fixed, predetermined routes are loaded directly into each router's routing table when the network boots up.
- **Analogy:** It functions like a fixed traffic light schedule that changes colors at exact intervals, completely ignoring whether the road is empty or heavily gridlocked.

Key Characteristics

- **Advantages:** It is highly simple to implement and introduces zero extra computational processing overhead on the routers while the network is running.
- **Disadvantages:** It is completely rigid; it cannot dynamically react to network congestion, broken links, or shifting traffic patterns.
- **Core Examples:** Shortest Path (when computed strictly offline), Flooding, and Flow-based routing.

2. Adaptive (Dynamic) Routing

Definition

Adaptive routing algorithms, also called dynamic routing, continuously change and optimize their routing decisions on the fly while the network is running.

How it Works

These algorithms constantly adjust their tables based on real-time feedback from the network, reacting to topology changes (like a link going down) or current traffic levels. Dynamic algorithms differ from one another based on three key parameters:

- **Information Source:** Where they collect their data from—some look only at local information at the router itself, some gather metrics from neighboring (adjacent) routers, and others collect data from all routers across the entire network.

- **Update Trigger (When routes change):** Routes can change at regular fixed time intervals (ΔT seconds), only when the traffic load shifts, or exclusively when the network topology changes.
- **Optimized Metric:** What specific parameter they are trying to optimize, such as physical distance, minimum number of hops, or estimated transit delay time.

Key Characteristics

- **Advantages:** Highly resilient; it automatically reroutes packets around failed links or heavily congested paths.
- **Disadvantages:** Introduces a much higher processing overhead because routers must constantly compute paths and exchange control packets.
- **Core Examples:** Distance Vector Routing, Link State Routing (OSPF), and Hierarchical routing.

Feature / Aspect	Static (Non-Adaptive) Routing	Dynamic (Adaptive) Routing
Decision Basis	Computed completely offline, in advance.	Computed continuously using live, current network conditions.
Reaction to Failures/Traffic	No, cannot adjust if a link drops or clogs.	Yes, automatically recalculates paths around problems.
Processing Overhead	Low; no ongoing computational cost.	Higher; requires active CPU cycles for continuous calculation.
Metrics Used	Fixed administrative weights or offline costs.	Dynamic metrics like link load, delay, bandwidth, or hop count.
Typical Protocols	Manual Static Routes, Flooding.	RIP (Distance Vector), OSPF / IS-IS (Link State).

What do you mean by the optimality principle and sink tree?

The **Optimality Principle** and **Sink Trees** are foundational theoretical concepts used as benchmarks to evaluate how effectively a routing algorithm determines the best paths in a network.

1. The Optimality Principle

Definition

The optimality principle is a general, mathematical rule regarding ideal (optimal) routing paths that remains true regardless of the network's physical layout (topology) or its current traffic conditions.

Core Statement

If router J lies on the optimal path from router I to router K , then the optimal path from J to K also follows that very same route.

Why It Matters

This principle acts as a strict standard or benchmark. Network engineers use it to verify whether a routing algorithm is genuinely producing the most efficient, mathematically optimal paths through a network. If a sub-path within a chosen route violates this principle, the overall path cannot be optimal.

2. Sink Trees

Definition

A **Sink Tree** is a collection of all the optimal routes from every single router in a network to one specific destination router.

When you gather these optimal paths and draw them together, they naturally form a tree structure where the destination router sits at the very base, or the "root".

Key Characteristics

- **Path Metrics:** The distance metric used to calculate a sink tree is very often measured simply by the number of hops (how many routers a packet must cross).
- **Not Always Unique:** A sink tree is not always unique to a destination. It is entirely possible for multiple, different trees to exist within the same network if they have the exact same total path lengths to the destination.
- **The Ultimate Goal:** The primary objective of all dynamic routing algorithms is to successfully discover and maintain efficient sink trees for every single router operating within the network.
- **Analogy:** Think of a sink tree like a blueprint of the "best roads home" from every neighborhood in a city. When you map all those optimal roads together, they form a tree shape that funnels traffic from every starting point toward that one final home destination.

What do you mean by routing metrics? Explain with figure and example.

A **routing metric** is a cost value assigned to a network path. A routing algorithm uses this value to choose the best path.

When a router has many paths to reach a destination, it adds up the costs of all links along each path. The router then chooses the path with the lowest total metric.

Different routing protocols use different types of metrics:

- **Hop Count (Path Length):** The total number of links or routers a packet must cross.
- **Delay:** The total time it takes for a packet to travel along the link in milliseconds.
- **Bandwidth:** The total data capacity of the link.
- **Load:** How busy or crowded a link currently is with traffic.
- **Reliability:** How safe the link is from failing or dropping data.

Consider the network topology above. A packet needs to travel from Router A to Router M. There are two possible paths:

Case 1: Using Hop Count as the Metric

If the protocol chooses paths based on Hop Count, it looks for the fewest routers:

- Path 1 (A → B → M): This path takes 2 hops.
- Path 2 (A → C → F → E → M): This path takes 4 hops.

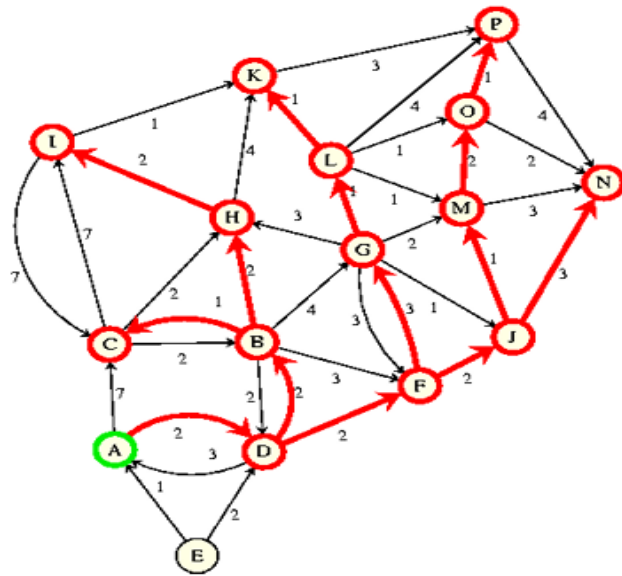
The Choice: The router chooses Path 1 because 2 hops is smaller than 4 hops.

Case 2: Using Delay as the Metric

If Path 1 is very crowded with background traffic, its links become slow. The Delay changes:

- Path 1 (A → B → M): High traffic makes the delay 50 ms per link. Total delay = 50 ms + 50 ms = 100 ms.
- Path 2 (A → C → F → E → M): High-speed fiber lines make the delay 10 ms per link. Total delay = 10 ms + 10 ms + 10 ms + 10 ms = 40 ms.

The Choice: The router chooses Path 2 because a total delay of 40 ms is faster than 100 ms.



Use shortest path algorithm to find shortest path from A to M

Step	Working Node	A	B	C	D	E	F	G	H	I	J	K	L	M
0	Initial	[0, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]
1	A	[0, -]	[∞, -]	[7, A]	[2, A]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]
2	D	[0, -]	[4, D]	[7, A]	[2, A]	[∞, -]	[4, D]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]
3	B	[0, -]	[4, D]	[5, B]	[2, A]	[∞, -]	[4, D]	[8, B]	[6, B]	[∞, -]	[∞, -]	[∞, -]	[∞, -]	[∞, -]
4	F	[0, -]	[4, D]	[5, B]	[2, A]	[∞, -]	[4, D]	[7, F]	[6, B]	[∞, -]	[6, F]	[∞, -]	[∞, -]	[∞, -]
5	C	[0, -]	[4, D]	[5, B]	[2, A]	[∞, -]	[4, D]	[7, F]	[6, B]	[12, C]	[6, F]	[∞, -]	[∞, -]	[∞, -]

6	H	[0, -]	[4, D]	[5, B]	[2, A]	[∞,-]	[4, D]	[7, F]	[6, B]	[8, H]	[6, F]	[10, H]	[∞,-]	[∞,-]
7	J	[0, -]	[4, D]	[5, B]	[2, A]	[∞,-]	[4, D]	[7, F]	[6, B]	[8, H]	[6, F]	[10, H]	[∞,-]	[7, J]

Final Result

- Shortest Path: $A \rightarrow D \rightarrow F \rightarrow J \rightarrow M$
- Total Minimum Cost: 7

State the steps of the shortest path algorithm.

Dijkstra's Shortest Path Algorithm Steps

- **Step 1: Initialize the Source Node**
 - Mark the starting (source) node as **permanent**. Its final shortest distance is fixed at **0**.
 - Set this source node as your current **working node**.
- **Step 2: Initialize All Other Nodes**
 - Set the tentative (temporary) distance to every other node in the network to **infinity** (∞).
- **Step 3: Update Neighbor Distances**
 - Look at all temporary neighbor nodes directly connected to your current working node.
 - Calculate a new distance to each neighbor:

New Distance = (Distance to working node) + (Link cost from working node to neighbor)
 - If this new distance is smaller than the neighbor's current distance, replace it with the new lower value. Record the working node as the path's previous step.
- **Step 4: Select the Next Working Node**
 - Look at all remaining temporary nodes across the entire network.
 - Choose the node with the **smallest** tentative distance.
 - Mark this chosen node as **permanent** and make it your new **working node**.
- **Step 5: Repeat Until Complete**
 - Repeat **Step 3** and **Step 4** until every single node in the graph is marked permanent. Once a node is permanent, its shortest path and distance from the source are final and will never change.

Explain with figure the count-to-infinity problem in DVR algorithm.

The **Count-to-Infinity problem** is a major weakness in Distance Vector Routing. It happens when a link or a router completely fails. Because routers only share data with their immediate neighbors, they cannot quickly see that a destination is now unreachable. Instead, they create a routing loop and slowly count upward toward a maximum value (infinity). This causes very slow convergence.

Consider five routers connected in a straight line topology: **A — B — C — D — E**.

Initially, all routers have correct paths to reach A. The hop count from each router (B, C, D, E) to A is 1, 2, 3, 4 respectively.

Step 1: The Link Fails

- The link between A and B breaks.
- B can no longer reach A directly. B sets its direct cost to A as infinity (∞).

Step 2: The Routing Exchange (The Mistake)

- At the next routing update, C shares its table with B. C claims: *"I can reach A in 2 hops."*
- B does not know that C's path to A actually goes through B itself.
- B thinks: *"I can reach C in 1 hop, and C can reach A in 2 hops, so I can reach A via C in $1 + 2 = 3$ hops."* B updates its table to 3 hops.

Step 3: The Upward Count

- On the next exchange, C sees that B now claims a distance of 3 hops to A.
- C updates its distance to: $1 + 3 = 4$ hops.
- Then, B sees C's new value and updates its distance to 5 hops.

Step 4: Reaching Infinity

This wrong loop continues slowly across all routers. The values keep climbing until they hit a predefined maximum number (for example, 16 in the RIP protocol). Only when it hits that maximum limit do the routers finally understand that A is completely unreachable.

Briefly explain the steps of Link State Routing.

Link State Routing is a dynamic routing algorithm where every router builds a complete map of the network topology and independently finds the shortest path.

Here are the five steps that every router performs:

The 5 Steps of Link State Routing

- **Step 1: Discover Neighbors**
 - When a router boots up, it sends a special **HELLO packet** out on its lines.
 - The routers at the other end reply with their addresses so everyone learns who their direct neighbors are.
- **Step 2: Measure Line Cost**
 - The router measures the delay (one-way transit time) to each neighbor by sending an **ECHO packet**.
 - The neighbor replies immediately. The router takes half of the round-trip time as the cost.
 - *Note:* Only transmission delay is measured; queue waiting time is ignored to prevent the paths from oscillating.
- **Step 3: Build Link State Packets (LSP)**
 - The router creates an LSP containing its identity, a sequence number, an age (TTL) value, and a list of its direct neighbors with their measured costs.
 - This packet is created at regular time intervals or whenever a major event happens (like a link going down).
- **Step 4: Distribute LSPs (Selective Flooding)**
 - The router sends its LSP to all other routers in the network using **selective flooding**.
 - Each router forwards received LSPs out on all lines except the one it arrived on. Routers use the sequence numbers to safely discard duplicate or old out-of-date packets.
- **Step 5: Compute New Routes**
 - Once a router collects LSPs from every router, it has a complete, identical map of the entire network topology.
 - The router runs **Dijkstra's algorithm** locally on this map to find the shortest path from itself to all destinations and updates its routing table.

Why do we need interconnecting networks and how to do that? How do the available networks differ from each other?

- **No Single Global Network:** No single network technology can physically or practically connect every computer in the world.
- **Diverse Local Technologies:** Different organizations, regions, and businesses build their own independent networks using different technologies that fit their specific needs.
- **The Goal of Internetworking:** We need internetworking to connect these separate, distinct networks together so they can function smoothly as one large, unified super-network (an internet). This allows any user on any network to communicate and share data with users on another network, regardless of the underlying technology they use.

When trying to connect different networks, engineers face the challenge that networks differ across many structural and operational layers. Key differences include:

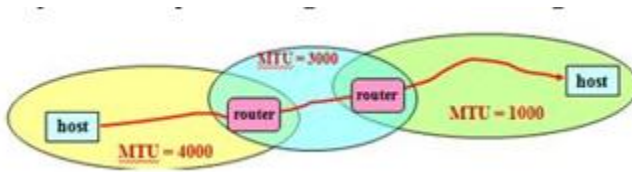
- **Services Offered:** Some networks use connection-oriented services, while others use connectionless services.
- **Protocols Used:** Networks run completely different sets of rules and protocols, such as IP, IPX, SNA, ATM, MPLS, or AppleTalk.
- **Addressing Schemes:** Networks vary in their address sizes, formats, and how those logical addresses are assigned to devices.
- **Packet Size Limits:** Different networks have different maximum packet sizes (Maximum Transmission Units or MTUs) that they can safely carry inside their frames.
- **Feature Support:** Some networks support broadcasting or multicasting features, while others completely lack these capabilities.
- **Reliability and Delivery Order:** Some networks guarantee reliable, in-order packet delivery, whereas others operate purely on a "best-effort" delivery basis where packets can arrive out of order or get dropped.

To connect different networks, specialized hardware components are used. These devices operate at different layers of the OSI reference model:

- **Repeaters and Hubs (Layer 1 - Physical Layer):** These devices simply amplify or copy electrical signals from one cable segment to another.
 - They do not read hardware or logical addresses. They are only used to extend the physical reach of a single network segment, not to connect truly different networks.
- **Bridges and Switches (Layer 2 - Data Link Layer):** These devices read and understand hardware MAC addresses.

- A switch intelligently forwards frames only to the specific port where the destination device resides. They can perform minor protocol translations (such as bridging an Ethernet segment to a Wi-Fi segment), but they still keep devices within a localized structure.
- **Routers (Layer 3 - Network Layer):** Routers are the true devices used for internetworking. They read logical IP addresses instead of hardware MAC addresses to make path decisions.
 - A router can successfully join completely different networks that use entirely different lower-layer standards and packet sizes. Multiprotocol routers can even handle multiple different network-layer protocols at the same time to ensure global communication.

Why do we need packet fragmentation? Do fragmentation for the given network.



Packet Fragmentation is needed because different networks have different **Maximum Transmission Unit (MTU)** sizes. If an IP packet is larger than the MTU of the next network, the router divides it into smaller fragments so that each fragment can pass through the network. Each fragment contains its own **20-byte IP header**.

Given

- Original IP Datagram = 4000 bytes
- IP Header = 20 bytes
- Data = 3980 bytes

Network 1 (MTU = 4000)

Since $4000 \leq 4000$, no fragmentation is needed.

Fragment	Header	Data	Total Length	MF	Offset
F1	20	3980	4000	0	0

Network 2 (MTU = 3000)

Maximum Data = MTU – Header

= $3000 - 20 = 2980$ bytes

Total Data = 3980 bytes

Fragment	Header	Data	Total Length	MF	Offset
F1	20	2980	3000	1	0
F2	20	1000	1020	0	2980

Network 3 (MTU = 1000)

Maximum Data = MTU – Header

= $1000 - 20 = 980$ bytes

Fragmenting F1 (Data = 2980 bytes)

Fragment	Header	Data	Total Length	MF	Offset
F1a	20	980	1000	1	0
F1b	20	980	1000	1	980
F1c	20	980	1000	1	1960
F1d	20	40	60	1	2940

Fragmenting F2 (Data = 1000 bytes)

Fragment	Header	Data	Total Length	MF	Offset
F2a	20	980	1000	1	2980
F2b	20	20	40	0	3960

Final Fragments

Final Fragment	Header	Data	Total Length	MF	Offset
F1a	20	980	1000	1	0
F1b	20	980	1000	1	980
F1c	20	980	1000	1	1960
F1d	20	40	60	1	2940
F2a	20	980	1000	1	2980
F2b	20	20	40	0	3960

Verification

Total Data:

$$980 + 980 + 980 + 40 + 980 + 20 = 3980 \text{ bytes}$$

This equals the original data size (**3980 bytes**), so the fragmentation is correct.

What are the fragmentation-related fields? Explain.

The 4 Fragmentation-Related Fields

1. Total Length Field

- **Purpose:** Indicates the total size of that specific, individual fragment.
- **Explanation:** It measures the size of the entire IP packet in bytes, which includes both the 20-byte IP header and the actual data payload carried inside that particular piece.

2. Identification Field

- **Purpose:** Groups related fragments together.
- **Explanation:** The sending host assigns a unique identification number to the original unsplit packet. When a router fragments that packet, it copies this exact same identification number into the header of every single piece. This allows the receiving destination to know which incoming fragments belong to the same original packet.

3. Flags Field

The flags field consists of control bits, two of which are critical for managing fragmentation:

- **Don't Fragment (DF) Flag:** If $DF = 1$, it explicitly instructs routers that they must **not** fragment the packet. If the packet reaches a link where it is too large to fit through the outgoing MTU, the router is forced to drop it and send an error message back to the sender. If $DF = 0$, routers are permitted to break the packet apart if needed.
- **More Fragments (MF) Flag:** This bit acts as a marker to let the receiver know if there are more pieces coming. If $MF = 1$, it means more fragments belonging to this packet will follow. If $MF = 0$, it indicates that this piece is the **last** (or only) fragment of the original packet, signaling the receiver to begin reassembly.

4. Fragment Offset Field

- **Purpose:** Identifies the correct position of the fragment data inside the original unsplit packet.
- **Explanation:** Because fragments can travel along different physical network paths, they often arrive at the final destination completely out of order. The offset field acts as a structural guide or index.
- **Important Rule:** In standard networking protocols, the offset value is measured strictly in units of **8-byte blocks** rather than individual bytes. For example, an offset value of 122 means the data payload in this fragment starts exactly at byte position $122 \times 8 = 976$ of the original payload data.

What are the network layer protocols? What are purposes of using an IP address?

Network Layer Protocols

The main network layer protocols are:

- **IP (Internet Protocol):** Delivers data from the source host to the destination host.
- **ARP (Address Resolution Protocol):** Finds the MAC address from an IP address.
- **RARP (Reverse Address Resolution Protocol):** Finds the IP address from a MAC address.
- **ICMP (Internet Control Message Protocol):** Reports network errors and sends status messages.
- **IGMP (Internet Group Management Protocol):** Manages multicast communication between hosts and routers.

Purposes of Using an IP Address

- Identifies every device uniquely on a network.
- Helps deliver data from the source to the correct destination.
- Supports **host-to-host** communication over different networks.
- Helps routers find the correct path for packet delivery.
- Provides logical addressing for communication across the Internet.

Why does the delivery of a packet need two levels of addressing (IP and MAC)?

Answer:

Packet delivery needs two levels of addressing because IP address and MAC address perform different jobs.

- IP address provides logical addressing and identifies the source and destination across different networks.
- MAC address provides physical addressing and delivers the packet to the correct device within the local network.
- A packet may travel through different physical networks, so IP is used for end-to-end communication, while MAC is used for local delivery at each hop.
- Therefore, both IP address and MAC address are necessary for successful packet delivery.

Explain static and dynamic mapping. State their limitations.

Answer:

Static Mapping

Static mapping is a method where the **IP address** and **MAC address** are stored in a fixed table on every computer. The mapping remains the same until it is changed manually.

Limitation:

- The table must be updated **manually** whenever the MAC address changes, making it difficult to manage.

Dynamic Mapping

Dynamic mapping is a method where the computer finds the address automatically when needed using **ARP**, **RARP**, or **DHCP**. It does not require a fixed address table.

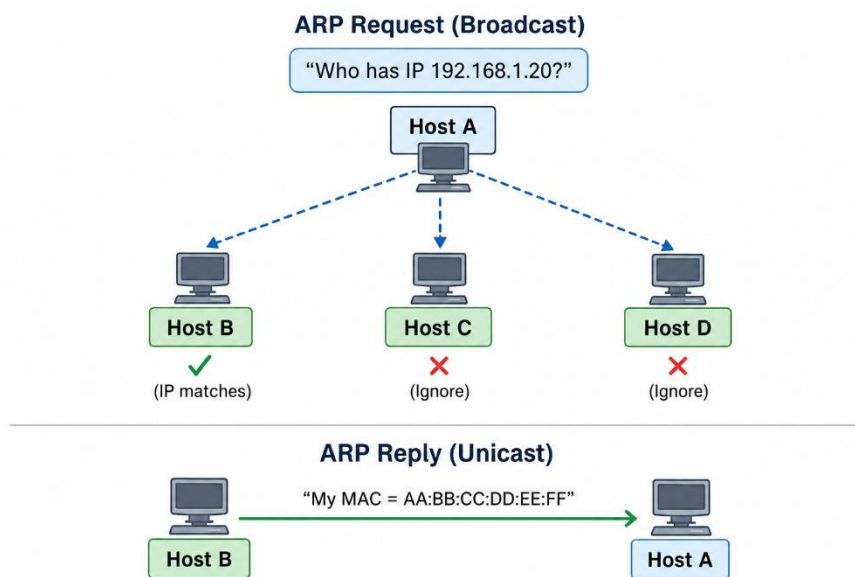
Limitation:

- It depends on network communication, so extra request and reply messages are needed before data transmission.

Show with figure that ARP request is broadcast and reply is unicast.

Answer:

When a host wants to know the MAC address of a destination, it sends an ARP Request as a broadcast to all devices in the local network. Only the device with the matching IP address sends an ARP Reply as a unicast directly to the sender.



Show ARP packet formation and encapsulation. What are the 4 cases of using ARP?

Answer:

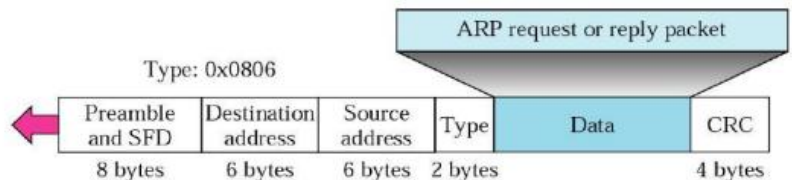
ARP Packet Formation

An **ARP packet** is used to map an **IP address** to a **MAC address**. It contains information about the sender, the target, and the operation type (**Request** or **Reply**).

Hardware Type		Protocol Type
Hardware length	Protocol length	Operation Request 1, Reply 2
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled in a request)		
Target protocol address (For example, 4 bytes for IP)		

ARP Encapsulation

Before transmission, the **ARP packet** is encapsulated inside an **Ethernet frame**. The **EtherType** field is set to **0x0806**, which indicates that the payload contains an ARP packet.



Four Cases of Using ARP

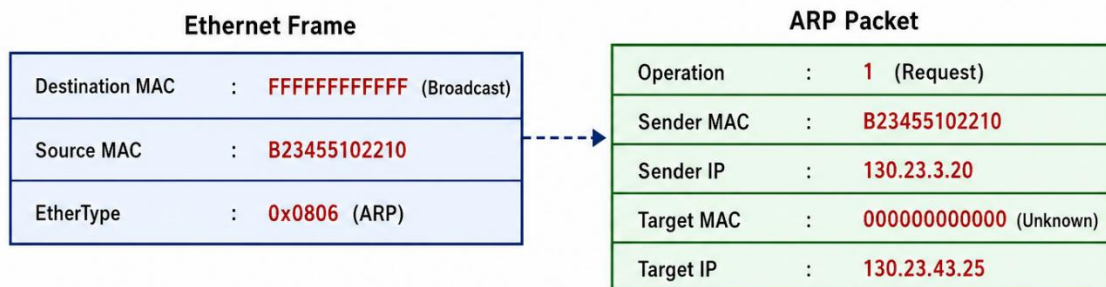
1. **Host → Host (Same Network):** ARP finds the destination host's **MAC address**.
2. **Host → Router (Different Network):** ARP finds the **MAC address** of the default router.
3. **Router → Router:** ARP finds the **MAC address** of the next-hop router.
4. **Router → Host:** ARP finds the destination host's **MAC address**.

A host with IP address 130.23.3.20 and physical address B23455102210 has a packet to send to another host with IP address 130.23.43.25 and physical address A46EF45983AB. The two hosts are on the same Ethernet network. Show the ARP request and reply packets encapsulated in Ethernet frames.

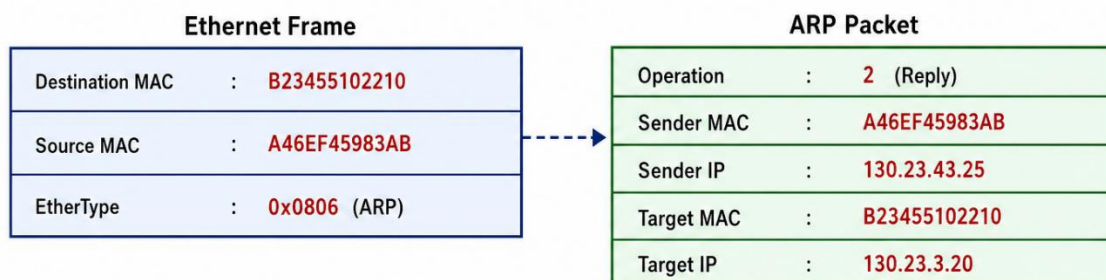
Answer:

Since both hosts are on the **same Ethernet network**, the sender first broadcasts an **ARP Request** to learn the destination's **MAC address**. The destination host then sends an **ARP Reply** directly to the sender with its MAC address. Both ARP packets are encapsulated inside Ethernet frames.

(a) ARP Request (Broadcast)



(b) ARP Reply (Unicast)



1. **ARP Request → Broadcast** (Destination MAC = FFFFFFFF)
2. **ARP Reply → Unicast** (Destination MAC = Sender's MAC)
3. **EtherType = 0x0806** indicates that the Ethernet frame carries an **ARP packet**.

Show RARP packet formation and encapsulation.

Answer:

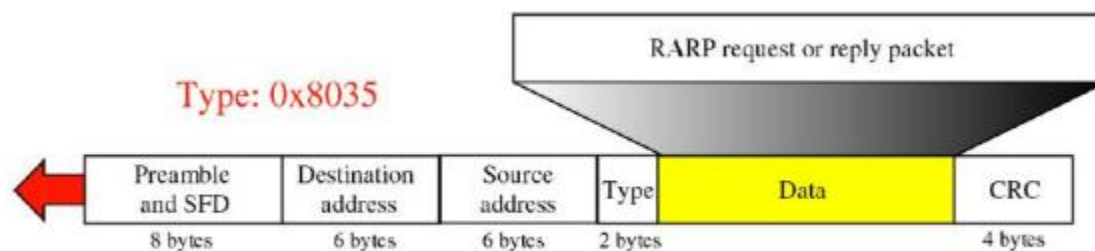
RARP Packet Formation

A RARP packet is used to find the IP address of a host from its MAC address. It has the same fields as an ARP packet, but the Operation field is 3 for Request and 4 for Reply.

Hardware type		Protocol type
Hardware length	Protocol length	Operation <i>Request 3, Reply 4</i>
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP) (It is not filled for request)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled for request)		
Target protocol address (For example, 4 bytes for IP) (It is not filled for request)		

RARP Encapsulation

Before transmission, the **RARP packet** is encapsulated inside an **Ethernet frame**. The **EtherType** field is **0x8035**, which indicates that the frame carries a RARP packet.



Explain using figure the four cases of using ARP.

Answer:

ARP is used to find the **MAC address** of the next device before sending data on a local network. There are **four cases** where ARP is used.

Case 1: Host → Host (Same Network)

Host A -----ARP-----> Host B

Finds Host B's MAC

Case 2: Host → Router (Different Network)

Host A -----ARP-----> Router

Finds Router's MAC

Case 3: Router → Router

Router 1 -----ARP-----> Router 2

Finds Next Router's MAC

Case 4: Router → Host

Router -----ARP-----> Host

Finds Host's MAC

What is the purpose of DHCP protocol and how does it work?

Answer:

Purpose of DHCP

DHCP (Dynamic Host Configuration Protocol) automatically assigns an **IP address** and other network settings, such as **subnet mask**, **default gateway**, and **DNS server**, to a host. It replaces **RARP** and reduces manual network configuration.

How DHCP Works

DHCP follows these **four steps**:

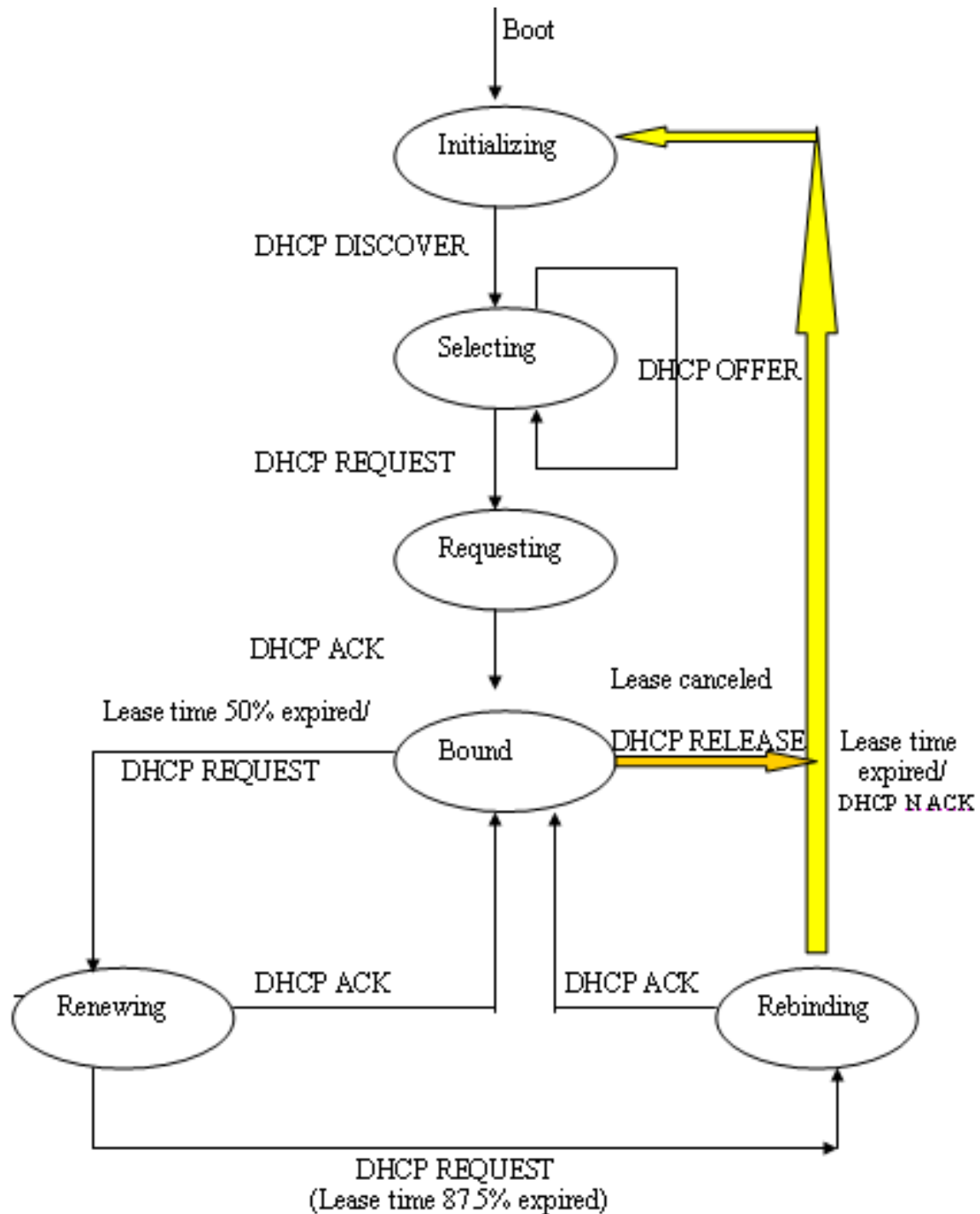
1. **DHCPDISCOVER:** The client broadcasts a request to find a DHCP server.
2. **DHCPOFFER:** The server offers an available IP address.
3. **DHCPREQUEST:** The client requests the offered IP address.
4. **DHCPACK:** The server confirms the request and assigns the IP address.

Draw state-transition diagram for DHCP. Explain all six states.

Answer:

A DHCP client passes through **six states** to obtain and maintain its IP address.

State-Transition Diagram



Six States

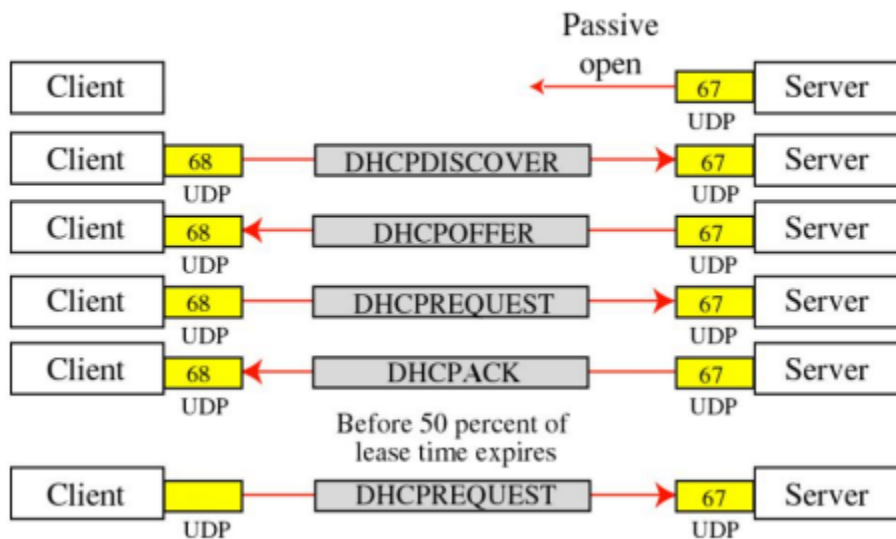
1. **Initializing:** The client sends **DHCPDISCOVER** to find a DHCP server.
2. **Selecting:** The client receives one or more **DHCPOFFER** messages and selects one server.
3. **Requesting:** The client sends **DHCPREQUEST** for the selected IP address.
4. **Bound:** The client receives **DHCPACK** and starts using the assigned IP address.
5. **Renewing:** At **50%** of the lease time, the client requests to renew the lease.
6. **Rebinding:** At **87.5%** of the lease time, if renewal fails, the client requests any DHCP server to renew the lease.

Show the DHCP message exchange between client-server.

Answer:

The DHCP client and server exchange **four messages** to assign an IP address.

(Exchanging messages)



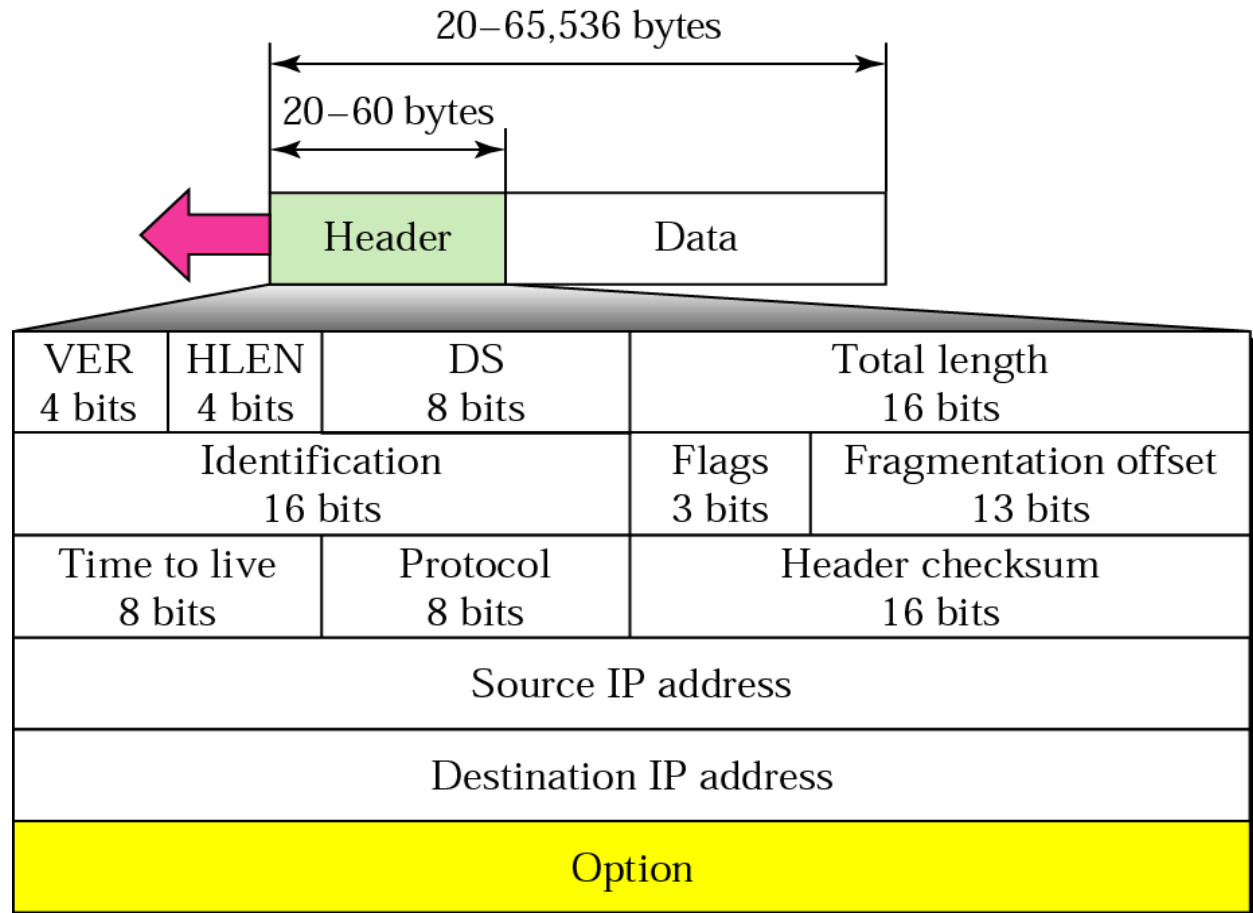
Message Description

- **DHCPDISCOVER:** Client searches for a DHCP server.
- **DHCPOFFER:** Server offers an available IP address.
- **DHCPREQUEST:** Client requests the offered IP address.
- **DHCPACK:** Server confirms the request and assigns the IP address.

Draw IPv4 and IPv6 datagram format and explain about each field briefly.

Answer:

(a) IPv4 Datagram Format

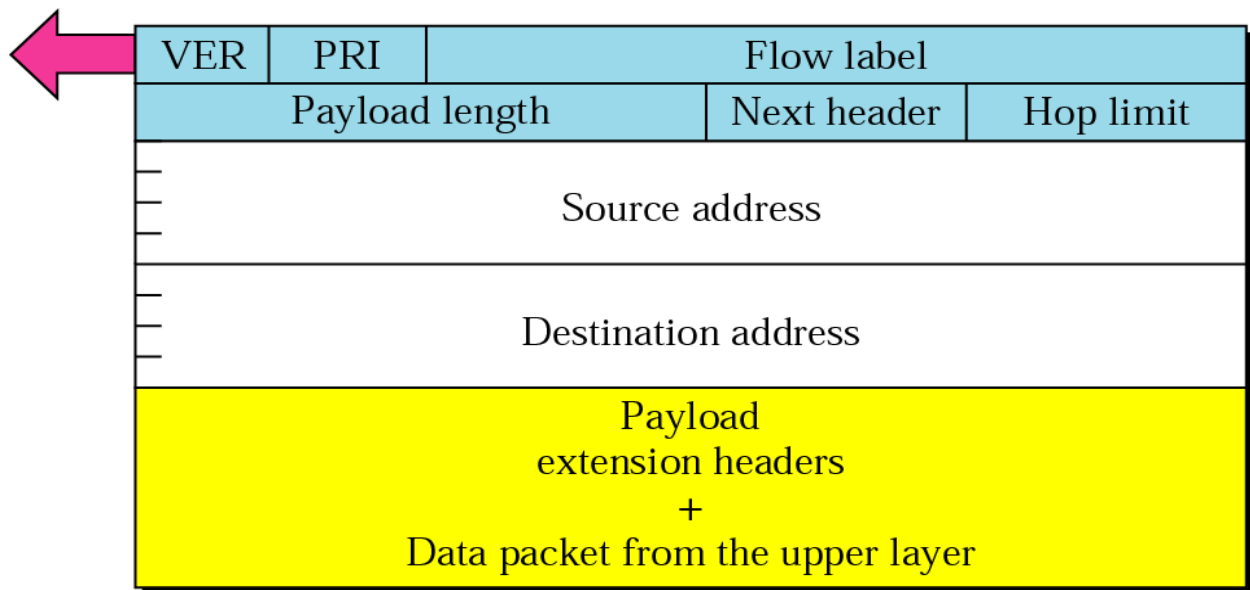


IPv4 Fields (Brief):

- **VER:** IP version (IPv4 = 4).
- **HLEN:** Header length.
- **Differentiated Services:** Provides Quality of Service (QoS).
- **Total Length:** Total size of the datagram.
- **Identification:** Identifies fragments of the same datagram.
- **Flags:** Controls fragmentation.
- **Fragment Offset:** Position of the fragment.
- **TTL:** Limits packet lifetime.
- **Protocol:** Identifies upper-layer protocol (TCP, UDP, etc.).

- **Header Checksum:** Detects header errors.
- **Source IP:** Sender's IP address.
- **Destination IP:** Receiver's IP address.
- **Options:** Optional control information.
- **Data:** Carries the actual message.

(b) IPv6 Datagram Format



IPv6 Fields (Brief):

- **VER:** IP version (IPv6 = 6).
- **Priority / Traffic Class:** Packet priority.
- **Flow Label:** Identifies a data flow.
- **Payload Length:** Size of payload.
- **Next Header:** Next protocol or extension header.
- **Hop Limit:** Limits packet lifetime.
- **Source Address:** Sender's IPv6 address.
- **Destination Address:** Receiver's IPv6 address.
- **Payload:** Actual data.

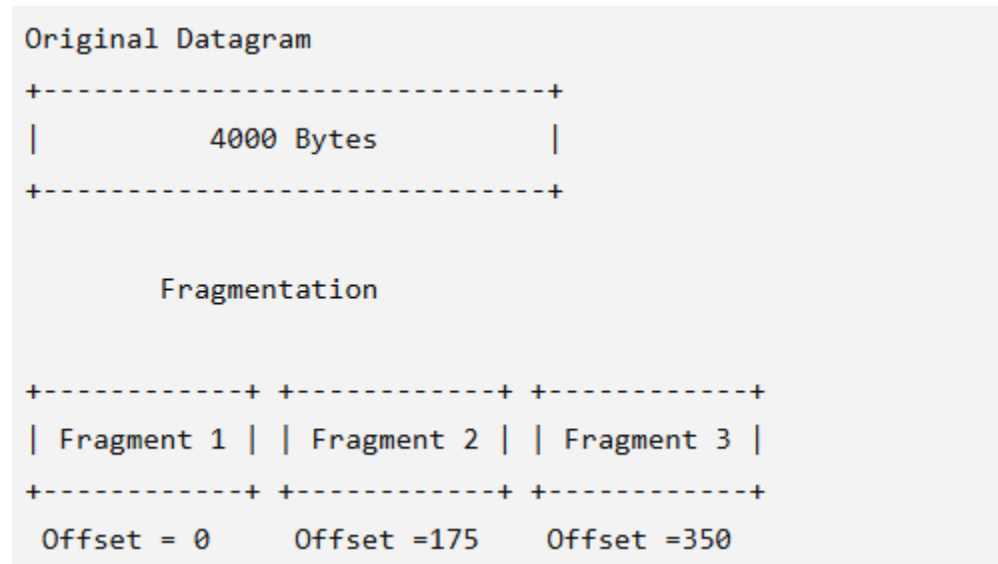
Define MTU and the fields of fragmentation with appropriate figure.

Answer:

MTU (Maximum Transfer Unit)

MTU (Maximum Transfer Unit) is the **maximum amount of data** that a network can carry in a single frame. If a packet is larger than the MTU, it is divided into **fragments**.

Figure



Fragmentation Fields

- **Identification:** Same value for all fragments of one datagram.
- **Flags:** Controls fragmentation.
 - **D (Do Not Fragment):** 1 = Do not fragment, 0 = Fragmentation allowed.
 - **M (More Fragments):** 1 = More fragments follow, 0 = Last fragment.
- **Fragment Offset:** Shows the position of the fragment in the original datagram (measured in 8-byte units).

What are the merits of IPv6 over IPv4?

Answer:

Merits of IPv6 over IPv4:

1. **Larger address space** with **128-bit** addresses.
2. **Simpler header** for faster processing.
3. **Supports extension headers** for new features.
4. **Built-in security** through authentication and encryption.
5. **Supports Flow Label** for better real-time communication.
6. **More flexible and scalable** for future network growth.

Explain three transition strategies from IPv4 to IPv6.

Answer:

During the transition from IPv4 to IPv6, three strategies are used to allow both protocols to work together.

1. Dual Stack

A device runs **IPv4 and IPv6** at the same time. It uses IPv4 or IPv6 depending on the destination address.

2. Tunneling

An **IPv6 packet** is encapsulated inside an **IPv4 packet** to travel through an IPv4 network.

3. Header Translation

The **IPv6 header** is converted into an **IPv4 header** (or vice versa) so that IPv4 and IPv6 devices can communicate.

Explain the fragmentation procedure of IPv6 packet and compare it to IPv4 fragmentation.

Answer:

IPv6 Fragmentation Procedure

In **IPv6**, **only the source host** can fragment a packet. Before sending, the source uses **Path MTU Discovery** to find the smallest MTU along the path. Routers do **not** fragment packets. If a packet is too large, the source host fragments it before transmission.

Comparison: IPv4 vs IPv6 Fragmentation

IPv4	IPv6
Source host and routers can fragment packets.	Only the source host can fragment packets.
Routers may fragment packets during transmission.	Routers never fragment packets.
No Path MTU Discovery is required.	Uses Path MTU Discovery before sending.

Explain Home address and Care-of address, Home agent and Foreign agent with appropriate figure.

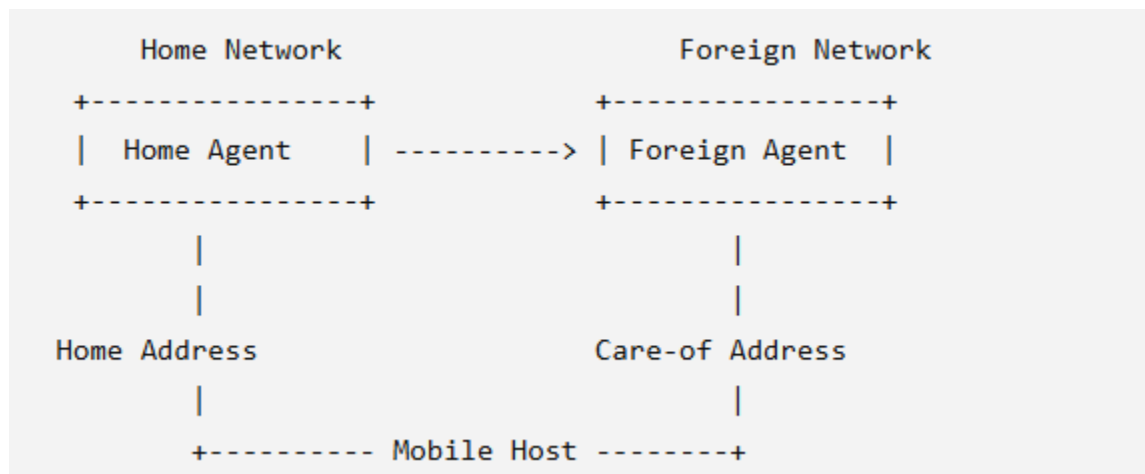
Answer:

In **Mobile IP**, a mobile host uses two addresses and two agents to maintain communication while moving between networks.

Definitions

- **Home Address:** The permanent IP address of the mobile host. It never changes.
- **Care-of Address:** A temporary IP address assigned in the visited network.
- **Home Agent (HA):** A router in the home network that keeps track of the mobile host and forwards packets.
- **Foreign Agent (FA):** A router in the visited network that receives forwarded packets and delivers them to the mobile host.

Figure

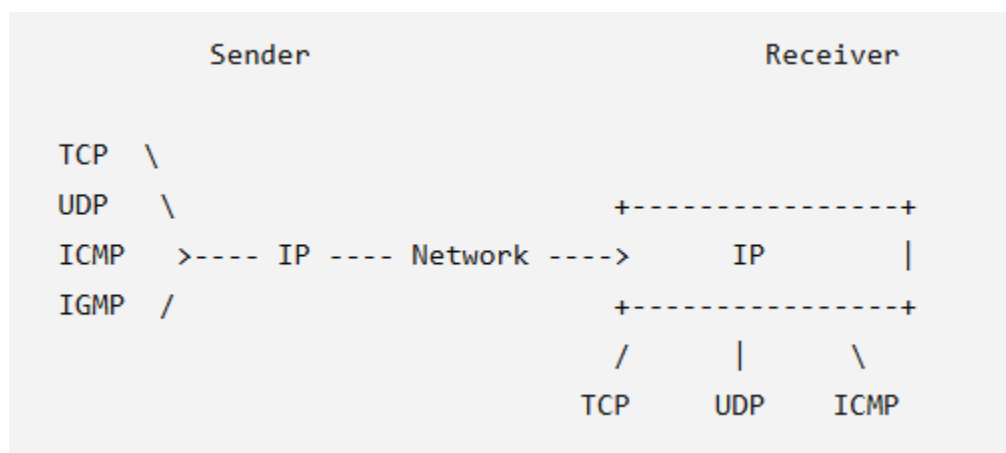


65. Explain multiplexing and de-multiplexing between network and transport layer using appropriate figure.

Answer:

Multiplexing combines data from different **transport layer protocols** into one **IP protocol** at the sender. **De-multiplexing** separates the received data and delivers it to the correct transport layer protocol at the receiver.

Figure

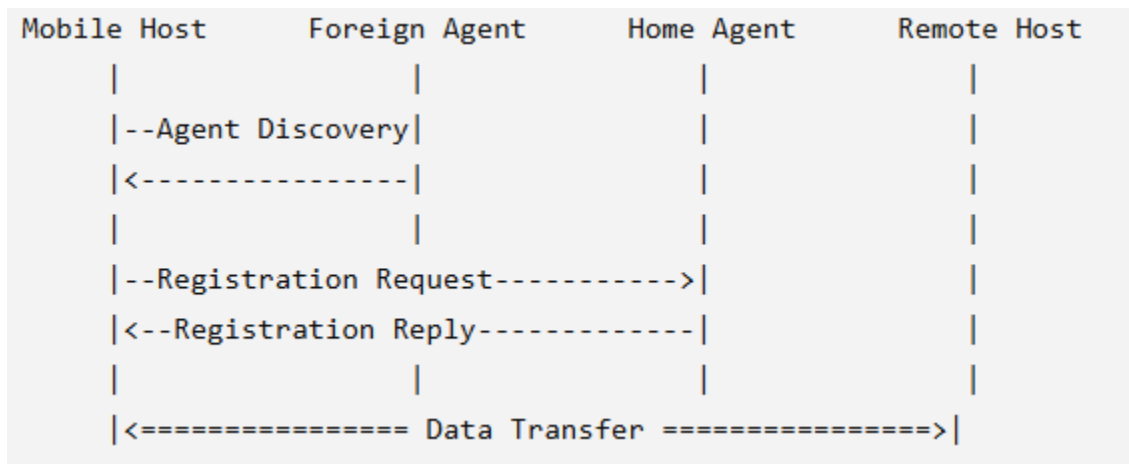


66. To communicate with a remote host, a mobile host goes through three phases: agent discovery, registration, and data transfer. Draw the time diagram.

Answer:

A mobile host communicates with a remote host through **three phases: Agent Discovery, Registration, and Data Transfer.**

Time Diagram



67. Communication involving Mobile IP can be inefficient. Explain the scenarios.

Answer:

Communication in **Mobile IP** may not always follow the shortest path. This causes two routing inefficiencies.

1. Double Crossing

When the remote host and mobile host are on the **same foreign network**, packets still travel through the **home agent** before reaching the mobile host. This causes unnecessary network traffic.

2. Triangle Routing (Dog-Leg Routing)

Packets travel from the **remote host → home agent → foreign agent → mobile host** instead of directly to the mobile host. This increases delay and reduces efficiency.

68. Differentiate between ATM cell and TCP packet.

Answer

ATM cells and TCP packets operate at different layers and handle data formatting differently:

Feature	ATM Cell PDF	TCP Packet / Segment PDF
OSI Layer	Data Link Layer (Layer 2)	Transport Layer (Layer 4)
Size Format	Fixed-size: Exactly 53 bytes	Variable-size: Can vary based on payload and MTU
Header Size	5 bytes	20 to 60 bytes
Data Switching	Switched using virtual circuits (VPI/VCI)	Routed using IP addresses and Port numbers
Primary Focus	Fast hardware switching in high-speed links	End-to-end reliable data delivery between processes

69. Describe both the ATM cell types with appropriate cell formation figure.

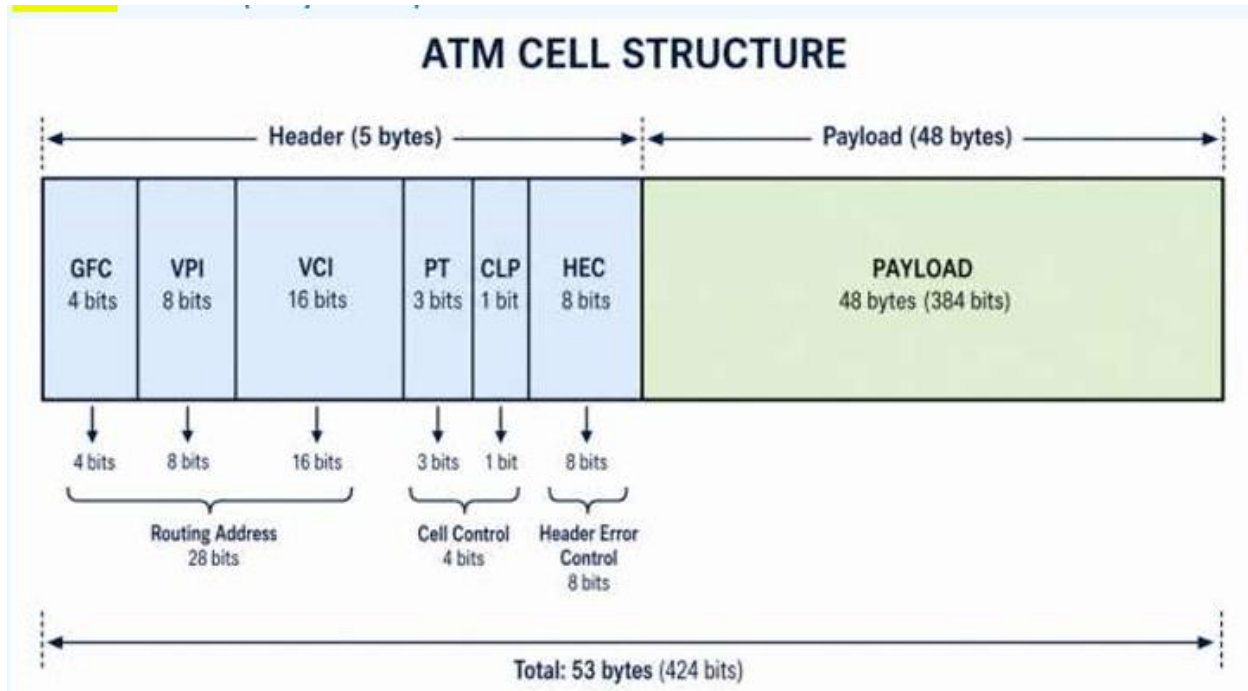
Answer

ATM defines **two** cell formats depending on where the connection is located:

1. **UNI (User-Network Interface):** Used between a user device (host) and an ATM switch.
2. **NNI (Network-Network Interface):** Used between two ATM switches inside the backbone network.

ATM Cell Structure (UNI Format):

An ATM cell consists of a **5-byte header** and a **48-byte payload**.



Field Description:

- **GFC (4 bits):** Generic Flow Control (present **only in UNI**, replaced by extra VPI bits in NNI).
- **VPI (8 bits in UNI / 12 bits in NNI):** Virtual Path Identifier.
- **VCI (16 bits):** Virtual Channel Identifier (VPI & VCI guide cells through switches).
- **PT (3 bits):** Payload Type (distinguishes user data from management cells).
- **CLP (1 bit):** Cell Loss Priority (if 1, cell can be dropped during congestion).
- **HEC (8 bits):** Header Error Control (CRC checksum for header only).

70. Write short note on ATM network and transport layer.

Answer

- **ATM Network Layer:** ATM provides a connection-oriented, cell-switched service. It uses Virtual Paths (VPI) and Virtual Channels (VCI) to establish Virtual Circuits (VC) before sending cells. This ensures predictable routing and Quality of Service (QoS) guarantees for voice, video, and data.
- **ATM Transport Layer (AAL - ATM Adaptation Layer):** Since higher-layer protocols send large variable-size packets (e.g., IP packets), the ATM Adaptation Layer acts as an interface between upper layers and the ATM layer. It segments variable-length packets into 48-byte cell payloads at the sender side and reassembles them back into packets at the receiver side.

71. How does TCP/UDP (transport layer protocol) works?

Answer

Transport layer protocols manage communication between processes running on different hosts using **Port Numbers**:

- **TCP (Transmission Control Protocol):**
 - Works in 3 steps: **Connection Setup (3-way handshake) → Data Transfer → Connection Termination.**
 - Ensures data arrives in order without errors using sequence numbers, acknowledgments (ACK), and retransmissions.
- **UDP (User Datagram Protocol):**
 - Operates on a **fire-and-forget (connectionless)** principle.
 - Takes data from the application layer, attaches port numbers and a simple checksum, and immediately sends it out without setting up a connection or waiting for ACKs.

72. Why would anyone use UDP? Clarify your answer with some applications where the UDP is used.

Answer

UDP is used because it provides **low latency, zero connection overhead, and fast speed**. Unlike TCP, UDP does not spend time establishing connections, tracking sequence numbers, or waiting for acknowledgments.

Why UDP is preferred in certain scenarios:

1. **Speed over Reliability:** Missing a single packet is better than delaying real-time data.
2. **Low Overhead:** Simple 8-byte header compared to TCP's 20-byte header.
3. **Supports Broadcast & Multicast:** Can send data to multiple receivers at once.

Applications using UDP:

- **Real-time Voice/Video Streaming:** Zoom, WhatsApp voice calls, YouTube live streams.
- **Online Gaming:** Fast player movements require low latency.
- **DNS (Domain Name System):** Requires small, single request-reply queries.
- **DHCP:** Used for quick IP assignment on bootup.

73. What is the difference between UDP and TCP internet protocols?

Answer

Feature	TCP	UDP
Connection Type	Connection-Oriented (Handshake required)	Connectionless (No handshake)
Reliability	Guaranteed delivery (Uses ACKs and retransmission)	Best-effort delivery (Packets can be lost)
Packet Order	Delivers packets in exact order	Packets may arrive out of order
Header Size	20 to 60 bytes	8 bytes
Flow/Congestion Control	Supported (Sliding Window, Slow Start)	Not supported
Speed	Slower (due to overhead)	Very Fast

74. What are the Challenges of Reliable Data Transfer?

Answer

Achieving reliable data transfer over an unreliable physical network poses several challenges:

1. **Packet Corruption:** Bits can be flipped or corrupted by noise in physical transmission media.
2. **Packet Loss:** Router queues can overflow during network congestion, dropping packets.
3. **Out-of-Order Delivery:** Packets taking different network paths may arrive out of sequence.
4. **Duplicate Packets:** Delayed acknowledgments can cause senders to retransmit packets that were already received.
5. **Flow Mismatch:** A fast sender can overwhelm a slow receiver's buffer space.

75. How TCP supports for TCP Support for Reliable Delivery?

Answer

TCP provides reliable data delivery over unreliable IP networks using five core mechanisms:

1. **Checksum:** Detects bit errors/corruption in headers and data.
2. **Sequence Numbers:** Keeps track of bytes sent so the receiver can reorder packets and detect duplicates.

3. **Acknowledgments (ACK):** Receiver sends ACK numbers back to inform the sender which bytes were received successfully.
4. **Timeouts & Retransmissions:** If an ACK is not received within a calculated RTT (Round Trip Time) window, TCP automatically retransmits the lost packet.
5. **Flow Control (Sliding Window):** Prevents the sender from overflowing the receiver's buffer by adjusting the window size based on receiver feedback.

Q112: Differentiate between 1G, 2G, 3G and 4G.

Answer

Mobile telephone systems evolved through four major generations:

Feature	1G PDF	2G PDF	3G PDF	4G PDF
Era	1980s	1990s	2000s	2010s
Technology	Analog	Digital	Digital	Digital (All-IP)
Services	Voice only	Voice + SMS	Voice + Mobile Internet	Broadband Internet + VoLTE
Modulation	FM	QPSK / TDMA	CDMA (W-CDMA)	OFDM
Data Speed	~2.4 kbps	~384 kbps	~2 Mbps	100+ Mbps
Example	AMPS	GSM	UMTS	LTE

Q113: Write short notes on IMPS and AMPS.

Answer

- **IMPS (Improved Mobile Telephone Service):** An early pre-cellular mobile telephone system introduced in the 1960s. It used a single high-power central transmitter covering an entire city with a limited number of channels, allowing very few simultaneous calls.
- **AMPS (Advanced Mobile Phone System):** The 1G analog cellular network standard introduced in the 1980s. It divided coverage areas into small hexagonal cells, allowing frequency reuse across different non-adjacent cells. It operated in the 800 MHz band with 832 full-duplex channels using Frequency Modulation (FM).

Q114: How AMPS accumulate more calls than IMPS?

Answer

AMPS accommodated significantly more calls than IMPS through two main mechanisms:

1. **Cellular Frequency Reuse:** Instead of using one big central tower covering an entire city (like IMPS), AMPS divided geographical areas into many small hexagonal cells. Non-adjacent cells could reuse the same frequency channels simultaneously without causing signal interference.
2. **Cell Splitting:** As the number of subscribers grew in a particular area, large cells were subdivided into smaller microcells. Each microcell reused frequencies independently, multiplying overall network capacity.

Q107. How to Measure the Network Performance?

Network performance means **how well a network works**. It is measured using the following factors:

- **Bandwidth:** The maximum amount of data the network can carry.
- **Throughput:** The actual amount of data transferred in a given time.
- **Latency (Delay):** The time taken for data to travel from sender to receiver.
- **Jitter:** The variation in packet delay.
- **Packet Loss:** The number of packets lost during transmission.
- **Reliability:** How stable and dependable the network is.

A network with **high throughput, low delay, low jitter, and low packet loss** has good performance.

Q142. Write Short Note on SNMP, MIBs, TRAP. What is the Purpose of MIBs?

SNMP (Simple Network Management Protocol)

SNMP is a protocol used to **monitor and manage network devices** such as routers, switches, and servers from a central location.

MIB (Management Information Base)

MIB is a **database** that stores information about network devices. The SNMP manager reads this information to monitor device status.

Purpose of MIB:

- Stores device information.
- Helps monitor network performance.
- Makes network management easier.

- Allows the manager to read or update device information.

TRAP

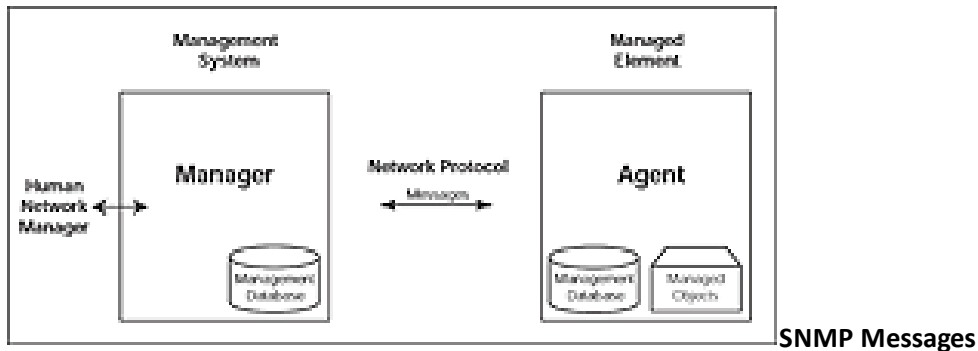
A **TRAP** is a message sent **automatically** by an SNMP agent to the SNMP manager when an important event or problem occurs.

Example: Device failure or link down.

Q143. Show SNMP Manager and SNMP Agent Communication Using SNMP Protocol. Explain Each Message.

SNMP works by exchanging messages between the **Manager** and the **Agent**.

Simple Diagram



- **Get Request:** Manager requests information from the agent.
- **Get Response:** Agent sends the requested information.
- **Set Request:** Manager changes or updates a value in the device.
- **Trap:** Agent automatically sends an alert when an important event happens.

Q111. Explain the Characteristics of Enterprise Network and Design Issues for this Network.

Characteristics of an Enterprise Network

An **enterprise network** is a network used by an organization or company to connect its employees, offices, and devices.

Characteristics:

- Connects many users and devices.
- High-speed communication.
- Secure and reliable.
- Easy to expand.

- Supports wired and wireless connections.
- Allows resource sharing.

Design Issues of an Enterprise Network

When designing an enterprise network, the following issues should be considered:

- **Scalability:** The network should support future growth.
- **Reliability:** The network should continue working even if one device fails.
- **Security:** Protect data from unauthorized access.
- **Performance:** Provide high speed with low delay.
- **Cost:** The network should be cost-effective.
- **Easy Management:** The network should be easy to monitor and maintain.